

12

MEASUREMENT AND SPACE

GEOMETRY AND NETWORKS

The word ‘geometry’ comes from the Greek word *geometria*, which means ‘earth measuring’. The principles and ideas of geometry and networks are evident in many aspects of our lives. For example, geometry can be seen in the design of buildings, bridges, roads and transport networks, such as this footbridge over the Yarra River in central Melbourne.



Chapter outline

	Working mathematically				
12.01 Angle sum of a polygon*	U	F		R	C
12.02 Transformations on the number plane*	U	F		R	C
12.03 Congruent triangles proofs*	U	F		R	C
12.04 Tests for quadrilaterals*	U	F	PS	R	C
12.05 Proving properties of triangles and quadrilaterals*	U	F		R	C
12.06 Formal geometrical proofs*	U	F	PS	R	C
12.07 Finding sides in similar figures	U	F	PS	R	C
12.08 Tests for similar triangles*	U	F		R	C
12.09 Similar triangles proofs*	U	F		R	C
12.10 Networks*	U	F		R	C
12.11 Planar graphs and Euler's formula*	U	F		R	C
12.12 Eulerian trails and circuits*	U	F		R	C

***PATH**

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Wordbank

congruence test One of 4 tests for proving that triangles are congruent: SSS, SAS, AAS and RHS

edge (of a network) A line that connects vertices (points) together

Eulerian trail A walk in a network in which every edge is used only once

included angle The angle between 2 given sides of a shape

planar graph A connected network (graph) where the edges do not cross (overlap)

quadrilateral test A property of a quadrilateral that proves that it is a particular type of quadrilateral, for example, if opposite angles are equal, then it *must* be a parallelogram

similarity test One of 4 tests for proving that triangles are similar: 'SSS', 'SAS', 'AA' and 'RHS'

vertex (of a network) A point representing an item that is connected by lines called edges

Quiz
Wordbank 12

Videos (11):

SkillCheck Geometry

12.03 Tests for congruent triangles • Congruent triangles proofs

12.04 Tests for quadrilaterals

12.05 Proving properties of a rectangle

12.07 Finding an unknown side in similar figures

12.08 Tests for similar triangles

12.09 Similar triangles proofs

12.10 Networks

12.11 Euler's formula for planar graphs

12.12 Eulerian trails

Twig videos (3):

12.07 The history of the golden ratio

12.10 The seven bridges of Konigsberg • Networks: Labyrinths and mazes

12.12 The seven bridges of Konigsberg

Quizzes (5):

- Wordbank 12
- SkillCheck 12
- Mental skills 12
- Language of maths 12
- Test yourself 12

Skillsheet (1):

12.07 Finding sides in similar triangles

Worksheets (18):

12.01 Angle sum of a polygon • Equal angles • Angles in polygons

12.02 Transformations • 5 mm grid paper

12.03 Congruent triangles proofs • Congruent triangles

12.04 Quadrilaterals: True or false?

12.05 Quadrilaterals: True or false? • Proving properties of quadrilaterals

12.07 A page of similar figures • Enlargements and reductions • Enlarging a logo • Finding sides in similar figures

12.08 Congruent and similar triangles proofs • Congruence and similarity review

12.09 Congruent and similar triangles proofs • Complete the proofs

12.12 Eulerian trails and circuits

Mind map Geometry and networks (Advanced)

Test yourself 12 5 mm grid paper

Puzzles (6):

SkillCheck Finding angles

12.06 Geometrical proof order activity

12.07 Cartoon enlargement • Similar triangles

Language of maths Geometry and networks crossword • Geometry crossword (Advanced)

Presentation (1):

12.05 Geometric problems and proofs

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

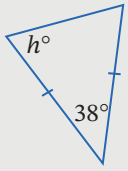
- (PATH) solve geometrical problems involving the angle sum of a polygon and the exterior angle sum of a convex polygon
- (PATH) translate, reflect and rotate points and shapes on a number plane using coordinates, where the reflections are across the x- or y-axis and the rotations through multiples of 90° about the origin
- (PATH) write formal proofs for congruent and similar triangles
- (PATH) identify and use the definitions and tests for the special quadrilaterals
- (PATH) investigate properties of special triangles and quadrilaterals using congruent triangles and solve related geometry problems
- (PATH) write formal geometrical proofs using logical reasoning

- understand the properties of similar figures
- use scale factors to find unknown sides in similar figures
- (PATH) describe networks using appropriate terminology including vertex, edge, face and degree
- (PATH) describe a planar graph and apply Euler's formula
- (PATH) describe connected graphs, walks, trails, circuits, paths, cycles
- (PATH) identify and investigate Eulerian trails and circuits

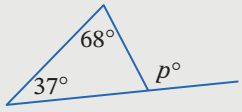
SkillCheck ANSWERS ON P. 670

1 Find the value of each variable.

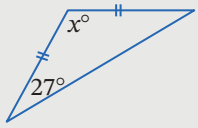
a



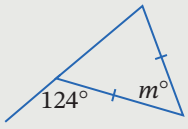
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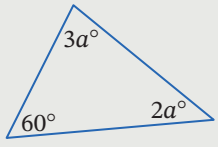
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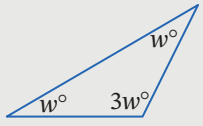
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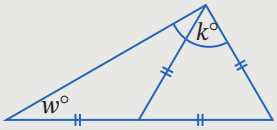
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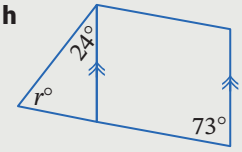
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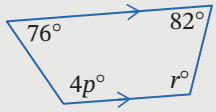
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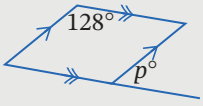
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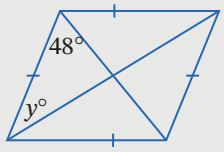
i



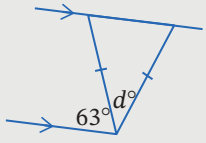
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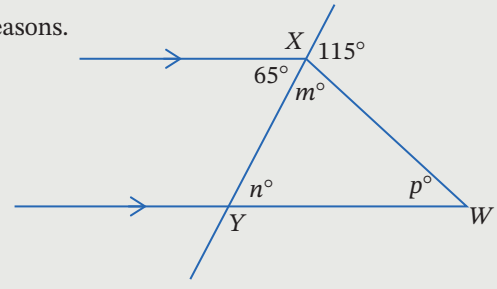


l



2 **a** Find the value of m, n *and* p, giving reasons.

b What type of triangle is $\triangle XYW$?



Quiz
SkillCheck 12

Puzzle
Finding angles

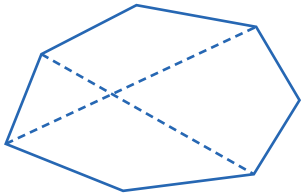
Video
Geometry

12.01 Angle sum of a polygon

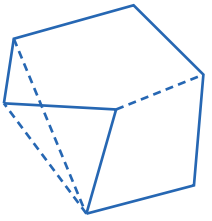
PATH

A **polygon** is any shape with straight sides.

Convex polygon



Non-convex polygon



Worksheets
Angle sum of a polygon
Equal angles
Angles in polygons

Angle sum of a polygon

The angle sum of a polygon with n sides is given by the formula

$$A = 180(n - 2)°.$$

This formula applies to both convex and non-convex polygons.

Example 1

Find the angle sum of a 15-sided polygon.

Solution

Angle sum $= 180\,(15 - 2)°$ $\quad n = 15$

$= (180 \times 13)°$

$= 2340°$

Example 2

Find the number of sides in a polygon that has an angle sum of 1080°.

Solution

$180\,(n - 2) = 1080$ $\quad A = 1080$

$180n - 360 = 1080$

$180n = 1440$

$n = \dfrac{1440}{180}$

$= 8$

$\therefore$ The polygon has 8 sides (octagon).

Angle in a regular polygon

The size of each angle in a **regular polygon** with n sides is:

$$\frac{\text{Angle sum}}{\text{Number of sides}} = \frac{180(n-2)°}{n}$$

Example 3

Find the size of one angle in a regular pentagon.

Solution

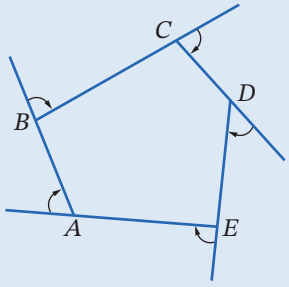
A pentagon has 5 sides ($n = 5$).

$$\text{Size of one angle} = \frac{180(5-2)^\circ}{5}$$
$$= \frac{(180 \times 3)^\circ}{5}$$
$$= 108^\circ$$

Each angle in a regular pentagon is 108°.

Exterior angle sum of a convex polygon

The sum of the exterior angles of a convex polygon is 360°.



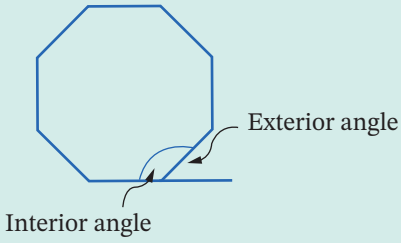
Worksheet
Angles in polygons

Example 4

For a regular octagon, find the size of:

a each exterior angle

b each (interior) angle.



Solution

a Sum of exterior angles = 360°

One exterior angle = 360° ÷ 8 = 45°

b Each angle = 180° − 45° (angles on a straight line)

= 135°

OR : Each angle $= \frac{180(8-2)^\circ}{8}$

= 135°

PATH

Example 5

Find the number of sides of a regular polygon if:

a each exterior angle is 24° **b** each (interior) angle is 140°.

Solution

a Number of exterior angles $= 360 \div 24$

$= 15$

$\therefore$ The regular polygon has 15 sides.

b Exterior angle $= 180° - 140°$ (angles on a straight line)

$= 40°$

Sum of exterior angles $= 360°$

Number of exterior angles $= 360° \div 40°$

$= 9$

$\therefore$ The regular polygon has 9 sides.

OR: $\frac{180(n-2)}{n} = 140$

$180(n-2) = 140n$

$180n - 360 = 140n$

$40n = 360$

$n = \frac{360}{40}$

$= 9$

EXERCISE 12.01 ANSWERS ON P. 670

Angle sum of a polygon UFRC

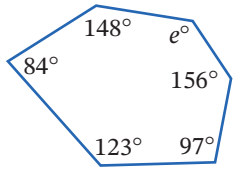
EXAMPLE 1

1 Find the angle sum of a polygon with:

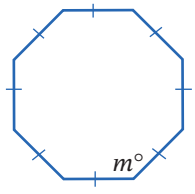
a 12 sides **b** 10 sides **c** 9 sides **d** 20 sides **e** 15 sides

2 Find the value of each variable. R

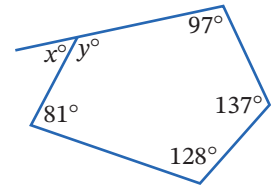
a



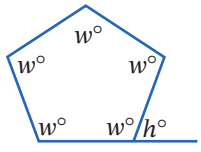
b



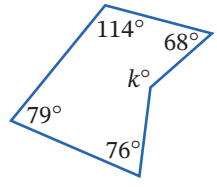
c



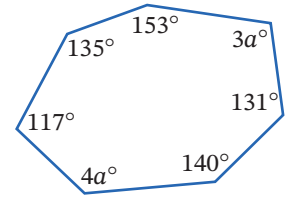
d



e



f



EXAMPLE 2

3 Find the number of sides in a polygon that has an angle sum of: R

a 720° **b** 3420° **c** 1980° **d** 5040° **e** 1260°.

4 The angle sum of a regular polygon is 2520°. R

a How many sides does the polygon have?

b Find the size of each angle.

Foundation Standard Complex

9780170479660

5 Find the size of one angle in a regular: R

a decagon **b** octagon **c** hexagon **d** dodecagon.

6 Find the size of one angle in a regular polygon with 15 sides. Select **A**, **B**, **C** or **D**. R

A 142° **B** 150.5° **C** 156° **D** 165°

7 How many sides does a regular polygon have if each of its angles is: R

a 168°? **b** 172°? **c** 165.6°?

8 Find the size of each exterior angle in a regular:

a pentagon **b** dodecagon **c** 18-sided polygon **d** hexagon.

9 Find the size of each angle in a regular: R

a nonagon **b** 20-sided polygon **c** decagon **d** 30-sided polygon.

10 Find the number of sides in a regular polygon if each exterior angle is:

a 15° **b** 72° **c** 20° **d** 40° **e** 5°

11 Find the number of sides in a regular polygon with an exterior angle of 12°.
Select the correct answer **A**, **B**, **C** or **D**.

A 12 **B** 15 **C** 30 **D** 45

12 Find the number of sides in a regular polygon if each angle is: R

a 135° **b** 144° **c** 156° **d** 178° **e** 165° **f** 150°.

PATH

EXAMPLE 3

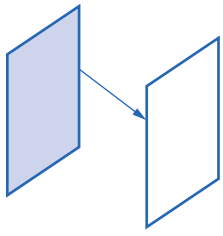
EXAMPLE 4

EXAMPLE 5

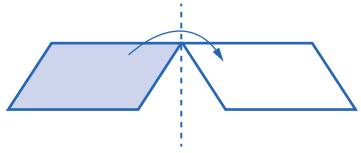
12.02

Transformations on the number plane 12.02

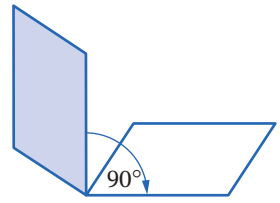
There are 3 types of **transformations**.



translation ('slide')



reflection ('flip')



rotation ('spin')

PATH

Worksheet
Transformations

The shapes have changed position (transformed) but still have the same shape and size.

The original shape is called the **original**.

The transformed shape is called the **image**.

When a point or vertex P of an original shape is transformed, the corresponding point or vertex on the image is labelled P', pronounced 'P-dash' or 'P-prime'.

Foundation Standard Complex

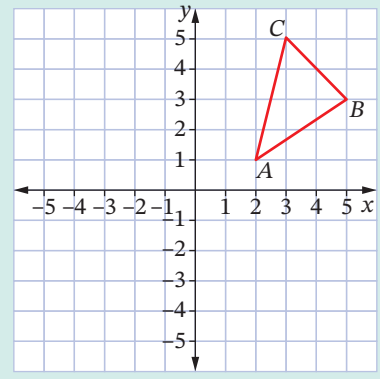
PATH

Example 6

a Reflect $\triangle ABC$ across the y-axis to obtain the image $\triangle A'B'C'$.

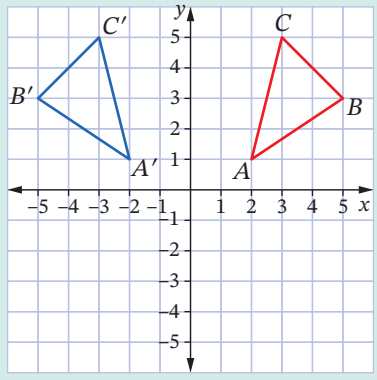
b Compare the coordinates of each vertex of $\triangle ABC$ to its matching vertex in $\triangle A'B'C'$.

c Now reflect $\triangle ABC$ across the x-axis to obtain the image $\triangle A'B'C'$. Compare the coordinates of each vertex of the original to its matching vertex in its image.



Solution

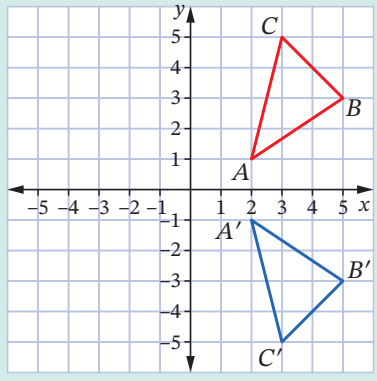
a



b $A(2, 1) \rightarrow A'(-2, 1)$, $B(5, 3) \rightarrow B'(-5, 3)$, $C(3, 5) \rightarrow C'(-3, 5)$

When reflected in the y-axis, the x-coordinate of each vertex changes sign while the y-coordinate stays the same.

c



$A(2, 1) \rightarrow A'(2, -1)$, $B(5, 3) \rightarrow B'(5, -3)$, $C(3, 5) \rightarrow C'(3, -5)$

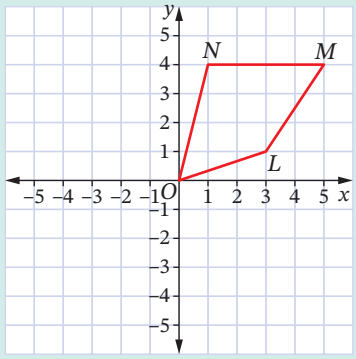
When reflected in the x-axis, the y-coordinate of each vertex changes sign while the x-coordinate stays the same.

Example 7

a Rotate $OLMN$ clockwise about O through each angle to obtain $OL'M'N'$:

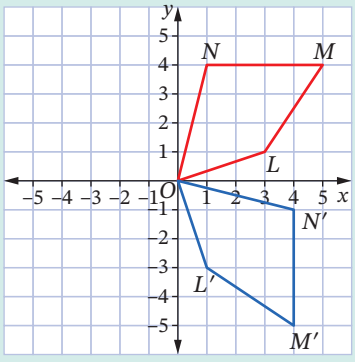
i 90° **ii** 180° **iii** 270°

b Compare the coordinates of each vertex of $OLMN$ to each vertex of the image $OL'M'N'$ for each of the rotations in part **a**.



Solution

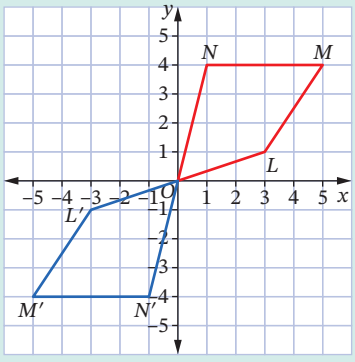
a i



b i $O(0, 0) \rightarrow O(0, 0)$. O is the centre of rotation.

$L(3, 1) \rightarrow L'(1, -3)$,
$M(5, 4) \rightarrow M'(4, -5)$,
$N(1, 4) \rightarrow N'(4, -1)$

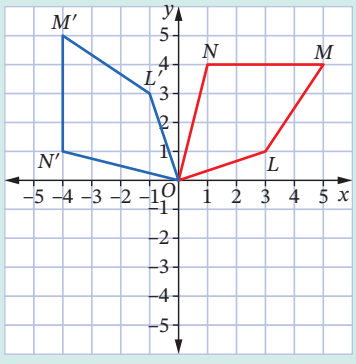
a ii



b ii $O(0, 0) \rightarrow O(0, 0)$,

$L(3, 1) \rightarrow L'(-3, -1)$,
$M(5, 4) \rightarrow M'(-5, -4)$,
$N(1, 4) \rightarrow N'(-1, -4)$

a iii



b iii $O(0, 0) \rightarrow O(0, 0)$

$L(3, 1) \rightarrow L'(-1, 3)$,
$M(5, 4) \rightarrow M'(-4, 5)$,
$N(1, 4) \rightarrow N'(-4, 1)$

PATH

Example 8

Find the new position of the point $K(-2, 5)$ if it is:

a rotated about the origin clockwise through 180°

b translated 4 units to the right and 7 units down

c reflected in the y-axis.

Solution

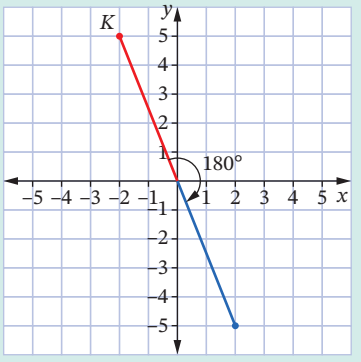
a K is rotated anticlockwise about the origin.

$\therefore$ the new position of K is $(2, -5)$

b The new position of K is $(-2 + 4, 5 - 7) = (2, -2)$

c K is reflected in the y-axis so the y-coordinate stays the same

$\therefore$ the new position of K is $(2, 5)$



Worksheet
5 mm grid paper

EXAMPLE 6

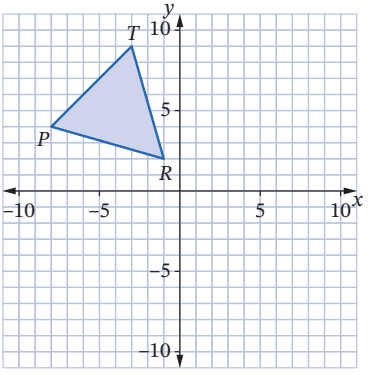
EXERCISE 12.02 ANSWERS ON P. 670

Transformations on the number plane U F R C

1 **a** Copy and reflect the shape $\triangle PRT$ across the y-axis to obtain the image $\triangle P'R'T'$.

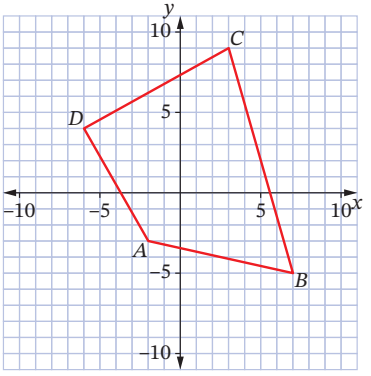
b Compare the coordinates of each vertex of $\triangle PRT$ to its matching vertex in $\triangle P'R'T'$.

c Now copy and reflect $\triangle PRT$ across the x-axis to obtain the image $\triangle P'R'T'$. Compare the coordinates of each vertex of $\triangle P'R'T'$ to its matching vertex in $\triangle PRT$. R C



2 **a** Copy and reflect the quadrilateral $ABCD$ across the x-axis to obtain the image $A'B'C'D'$.

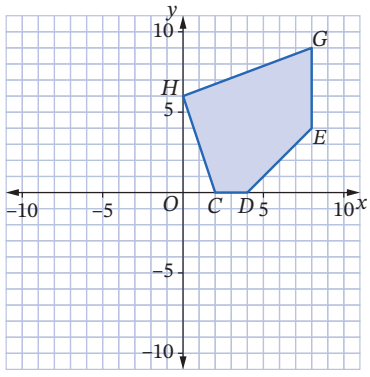
b Write the coordinates of each vertex of $A'B'C'D'$ and compare them to the coordinates of the vertices of $ABCD$. R C



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3 **a** Copy and rotate the shape $CDEGH$ clockwise about O through 90º to create the image $C'D'E'G'H'$.



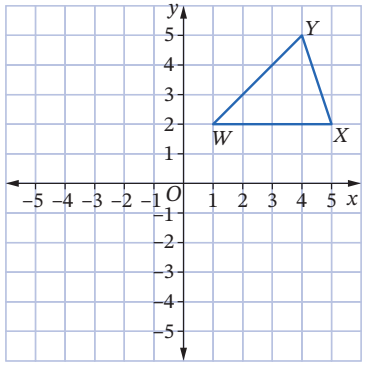
b Write down the coordinates of D' and E' and compare them to the coordinates of D and E.

c Write down the new positions of C, G and H if $CDEGH$ is rotated about O through: R C

i 180º **ii** 270°

4 **a** Copy and rotate $\triangle WXY$ clockwise about O through the following angles to obtain $\triangle W'X'Y'$:

i 90º **ii** 180° **iii** 270°



b Compare the coordinates of the vertices of $\triangle WXY$ to those of $\triangle W'X'Y'$ for each of the rotations. R C

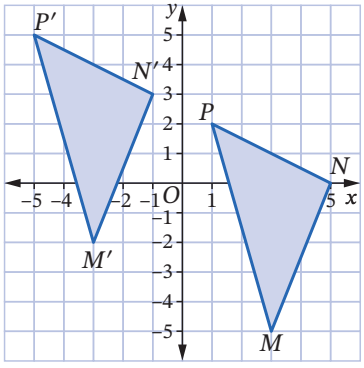
c Copy and rotate $\triangle WXY$ anticlockwise about O through the following angles

i 90° **ii** 180° **iii** 270°

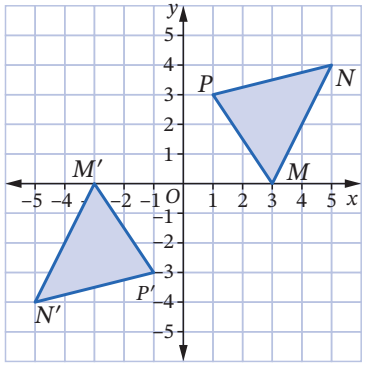
d Which of the images from part **c** are the same as the images from part **a**?

5 Describe the transformations that has been performed on $\triangle MNP$ in each diagram. R C

a



b

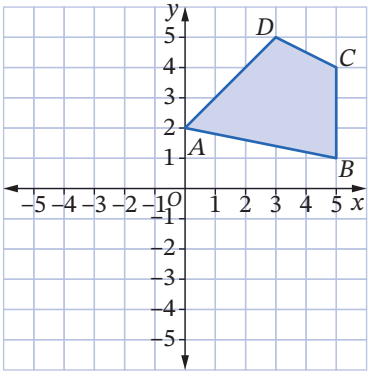


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PATH

6 **a** Rotate the quadrilateral $ABCD$ about O through 270° and then reflect the image in the x-axis to obtain $A'B'C'D'$.

b Compare the coordinates of the vertices of $ABCD$ to those of $A'B'C'D'$. R C



EXAMPLE 8

7 The point $W(-2, -5)$ is rotated through 180° about the origin to obtain the point W'.
What are the coordinates of W'? Select **A**, **B**, **C** or **D**.

A (5, 2) **B** (2, 5) **C** (−5, 2) **D** (−2, 5)

8 Find the coordinates of the image point A' if $A(-4, -6)$ is:

a reflected in the x-axis **b** reflected in the y-axis

c translated 4 units right and 7 units up

d rotated clockwise about the origin through

i 90° **ii** 180° **iii** 270°

9 What is the new position of $D(3, -8)$ if it is reflected in the x-axis and then rotated clockwise about the origin through 90°? Select **A**, **B**, **C** or **D**.

A (8, −3) **B** (−3, 8) **C** (−8, 3) **D** (−3, −8)

10 Find the new position of $X(-3, -7)$ if it is: R C

a rotated through 180° about the origin

b translated 5 units to the right and 9 units up

c reflected in the x-axis

d rotated clockwise through 90° and then reflected in the y-axis

e translated 6 units to the right, 4 units up and then rotated anticlockwise about the origin through 90°

11 The point $T(3, -4)$ is the image of a point after it has rotated clockwise through 180° and then has been translated 3 units down and 5 units to the right.
Find the original point. R C

12 Find the original point if (1, 5) is the image of a point that has been rotated 90° about the origin and then reflected in the x-axis. There are 2 possible answers.

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Technology

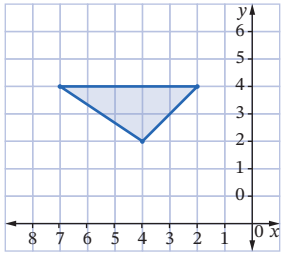
Transformations on the number plane

In this activity you will create reflections, rotations and translations of various shapes using **dynamic geometry software.** Create a number plane that shows all 4 quadrants and values of x and y from –10 to 10.

Activity 1: Reflection

1 Draw a **triangle** with vertices at (–7, 5), (–2, 5) and (–4, 3).

2 Draw a **reflection line on the *y*-axis,** through the points (0, 6) and (0, 0).



3 Reflect the triangle across the line.

4 Click a vertex and drag it. What happens?

5 Draw a new **reflection line** through the points (2, 7) and (–5, –2).

6 Reflect the triangle across this line.

Activity 2: Rotation

1 Draw a **polygon** with vertices at (0, 3), (–2, 3), (–3, 2), (–2, 0) and (0, 3).

2 Select the polygon and rotate it 45° clockwise about the point (0, 3).

3 Repeat step **2**, then rotate the polygon 180° clockwise about (0, 3). What happens?

4 Try other points as the centre point and enter different angles. What happens?

Activity 3: Translation

1 Draw a **polygon** with vertices at (–3, 4), (–3, –1), (1, –1), (1, 1), (–2, 1), (–2, 4) and (–3, 4).

2 Translate the polygon so that the image of (1, 1) is (4, 3).

3 Now translate the shape in 2 other ways.

Activity 4

1 Construct a **Polygon**.

2 Reflect the polygon about a point.

3 Reflect the polygon about a line.

4 Rotate the polygon by an angle.

5 Translate the polygon by a vector.

12.03 Congruent triangles proofs

PATH

Worksheets
Congruent triangles proofs

Congruent triangles

Videos
Tests for congruent triangles

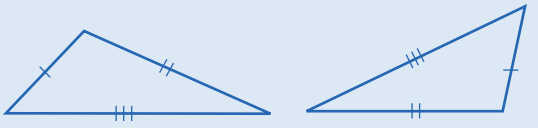
Congruent triangles proofs

Congruence tests

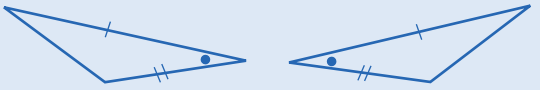
There are 4 tests for congruent triangles: **SSS**, **SAS**, **AAS** or **RHS**.

Two triangles are congruent if:

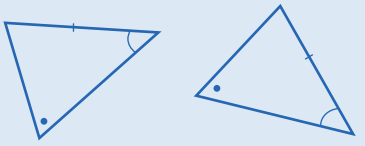
- the 3 sides of one triangle are respectively equal to the 3 sides of the other triangle (**SSS** rule)



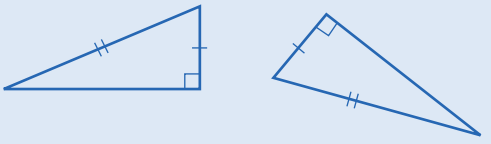
- 2 sides and the **included angle** of one triangle are respectively equal to 2 sides and the **included angle** of the other triangle (**SAS** rule)



- 2 angles and one side of one triangle are respectively equal to 2 angles and the matching side of the other triangle (**AAS** rule)



- they are right-angled and the hypotenuse and another side of one triangle are respectively equal to the hypotenuse and another side of the other triangle (**RHS** rule).

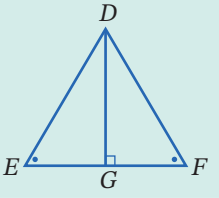


$\triangle ABC \equiv \triangle XYZ$ also means $\angle A = \angle X$, $\angle B = \angle Y$, $\angle C = \angle Z$.

Video
Congruent triangles proofs

Example 9

In this diagram, $\angle DEF = \angle DFE$ and $DG \perp EF$.



a Prove that $\triangle DEG \equiv \triangle DFG$.

b Hence show that G is the midpoint of EF.

Solution

a In $\triangle DEG$ and $\triangle DFG$:

Identifying the triangles in matching order of vertices

$\angle DEF = \angle DFE$ (given)

$\angle DGE = \angle DGF = 90°$ ($DG \perp EF$)

Stating each part of the congruence test, giving reasons

DG is common

$\therefore \triangle DEG \equiv \triangle DFG$ (AAS)

Concluding the congruence proof, stating the test used

b $\therefore EG = GF$ (matching sides of congruent triangles)

$\therefore G$ is the midpoint of EF.

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Example 10

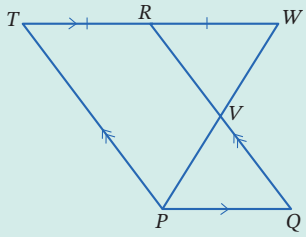
In the diagram, $PQRT$ is a parallelogram.

TR is extended to W so that $TR = RW$.

Prove that:

a $\triangle PQV \equiv \triangle WRV$

b V is the midpoint of PW.



Solution

a In $\triangle PQV$ and $\triangle WRV$:

$\angle Q = \angle WRV$ (alternate angles, $PQ \parallel TW$)

$\angle PVQ = \angle WVR$ (vertically opposite angles)

$PQ = TR$ (opposite sides of a parallelogram equal)

$TR = RW$ (given)

$\therefore PQ = RW$

$\therefore \triangle PQV \equiv \triangle WRV$ (AAS)

b $\therefore PV = WV$ (matching sides of congruent triangles)

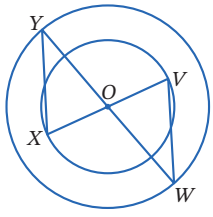
$\therefore V$ is the midpoint of PW.

EXERCISE 12.03 ANSWERS ON P. 672

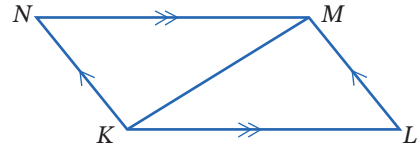
Congruent triangles proofs U F R C

EXAMPLE 9

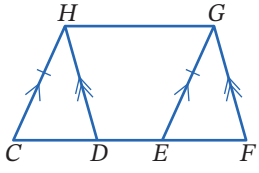
1 O is the centre of both circles. Prove that $\triangle XOY \equiv \triangle VOW$. R C



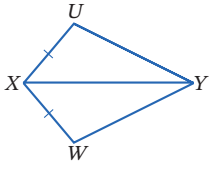
2 $KLMN$ is a quadrilateral with opposite sides parallel. Prove that $\triangle KLM \equiv \triangle MNK$. R C



3 $CH \parallel EG$, $DH \parallel$ FG and $CH = EG$. Prove that $\triangle CDH \equiv \triangle EFG$. R C



4 YX bisects $\angle UXW$ and $UX = WX$. Prove that $\triangle UXY \equiv \triangle WXY$. R C

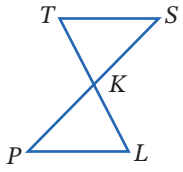


Foundation | Standard | Complex

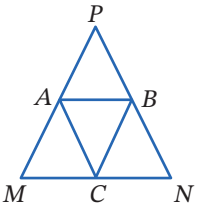
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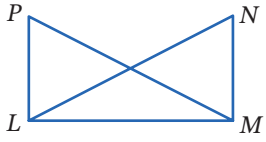
5 K is the midpoint of TL and SP. Prove that $\triangle TSK \equiv \triangle LPK$. R C



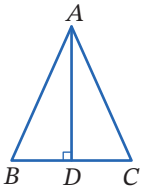
6 $PM = PN$ in this isosceles triangle and A, B and C are the midpoints of PM, PN and MN. Prove that $\triangle AMC \equiv \triangle BNC$. R C



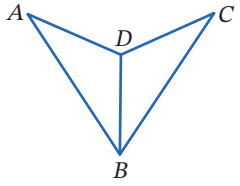
7 If $PL \perp LM$, $NM \perp LM$ and $PL = NM$, prove that $\triangle PLM \equiv \triangle NML$. R C



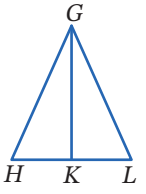
8 In $\triangle ABC$, $AB = AC$ and $AD \perp BC$. Prove that $\triangle ABD \equiv \triangle ACD$ and hence AD bisects $\angle BAC$. R C



9 $AB = CB$ and DB bisects $\angle ABC$. Prove that $\triangle ABD \equiv \triangle CBD$. R C



10 If K is the midpoint of HL and $GK \perp HL$, prove that $\triangle HKG \equiv \triangle LKG$. R C

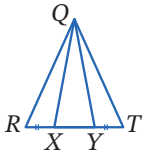


EXAMPLE 10

11 $\triangle QXY$ is isosceles with $QX = QY$. If $RX = TY$, prove that:

a $\triangle QRX \equiv \triangle QTY$

b $\triangle QRT$ is isosceles.

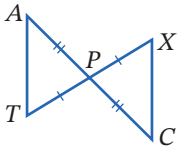


12 $TP = XP$ and $AP = CP$. Prove that:

a $\triangle TAP \equiv \triangle XCP$

b $TA = XC$

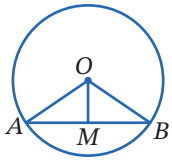
c $TA \parallel XC$



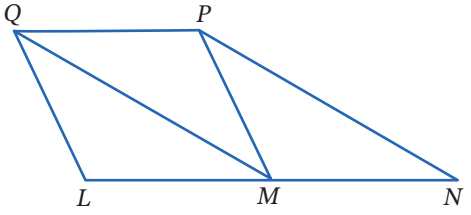
13 O is the centre of the circle and OM bisects AB. Prove that: R C

a $\triangle OAM \equiv \triangle OBM$

b OM bisects $\angle AOB$.



14 $LMPQ$ is a parallelogram. If M is the midpoint of LN prove that $\triangle LQM \equiv \triangle MPN$. R C



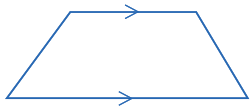
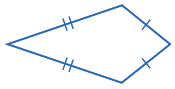
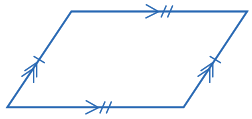
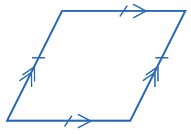
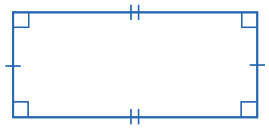
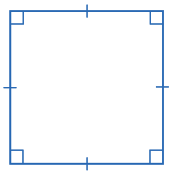
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Tests for quadrilaterals

12.04

The special quadrilaterals

Quadrilateral	Formal definition	Properties
Trapezium 	A quadrilateral with at least one pair of opposite sides parallel.	None
Kite 	A convex quadrilateral with 2 pairs of equal adjacent sides.	• One pair of opposite angles equal • Diagonals intersect at right angles
Parallelogram 	A quadrilateral with both pairs of opposite sides parallel.	• Opposite sides equal • Opposite angles equal • Diagonals bisect each other
Rhombus 	A parallelogram with 2 adjacent sides equal in length.	• All sides equal • Diagonals bisect each other at right angles • Diagonals bisect the angles of the rhombus
Rectangle 	A parallelogram with one angle a right angle.	• All angles are right angles • Diagonals are equal in length
Square 	A rectangle with 2 adjacent sides equal in length.	• All sides equal • All angles are right angles • Diagonals are equal in length • Diagonals bisect each other at right angles • Diagonals bisect the angles of a square

PATH



Worksheet
Quadrilaterals: True or false?

12.04

Note that the definitions of the quadrilaterals give *minimum conditions*. For example, a rhombus is defined as 'a parallelogram with 2 adjacent sides equal in length'. However, opposite sides are equal in a parallelogram so that means the rhombus has *all* sides equal in length.

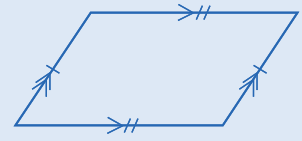
Some properties of the quadrilaterals can be used as minimum conditions to prove or test that a quadrilateral is a parallelogram, rectangle, square or rhombus. For example, if opposite angles are equal, then it *must* be a parallelogram.

PATH

Tests for quadrilaterals

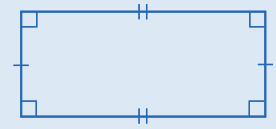
A quadrilateral is a **parallelogram** if one of these conditions is true:

- both pairs of opposite angles are equal, or
- both pairs of opposite sides are equal, or
- both pairs of opposite sides are parallel, or
- one pair of opposite sides are equal and parallel, or
- the diagonals bisect each other.



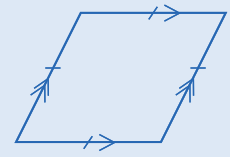
A quadrilateral is a **rectangle** if one of these conditions is true:

- all angles are 90°, or
- diagonals are equal and bisect each other.



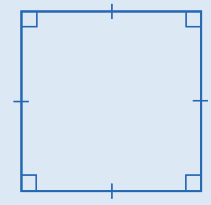
A quadrilateral is a **rhombus** if one of these conditions is true:

- all sides are equal, or
- diagonals bisect each other at right angles.



A quadrilateral is a **square** if one of these conditions is true:

- all sides are equal and one angle is 90°, or
- all angles are 90° and 2 adjacent sides are equal, or
- diagonals are equal and bisect each other at right angles.

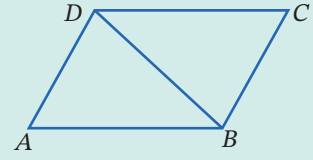


Video
Tests for quadrilaterals

Example 11

ABCD is a quadrilateral with *BD* as a diagonal.

Prove that if the opposite sides of *ABCD* are equal, then it must be a parallelogram.



Solution

In $\triangle ABD$ and $\triangle CDB$:

$AD = CB$	(opposite sides of *ABCD* are equal)
$AB = CD$	(opposite sides of *ABCD* are equal)
BD is common	
$\therefore \triangle ABD \equiv \triangle CDB$	(SSS)
$\therefore \angle ABD = \angle CDB$	(matching angles of congruent triangles)
$\therefore AB \parallel CD$	(alternate angles are equal)
Also, $\angle ADB = \angle CBD$	(matching angles of congruent triangles)
$\therefore AD \parallel CB$	(alternate angles are equal)
$\therefore$ *ABCD* is a parallelogram	(opposite sides are parallel)

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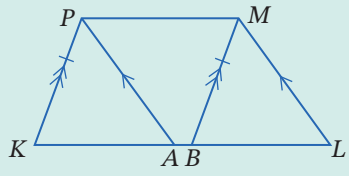
Example 12

In the diagram, $KP \parallel BM$, $AP \parallel LM$ and $KP = BM$.

Prove that:

a $\triangle KAP \equiv \triangle BLM$

b $ALMP$ is a parallelogram



Solution

a In $\triangle KAP$ and $\triangle BLM$:

$\angle PKA = \angle MBL$	(corresponding angles, $KP \parallel BM$)
$\angle KAP = \angle BLM$	(corresponding angles, $AP \parallel LM$)
$KP = BM$	(given)
$\therefore \triangle KAP \equiv \triangle BLM$	(AAS)

b $\therefore AP = LM$ (matching sides of congruent triangles)

$AP \parallel LM$ (given)

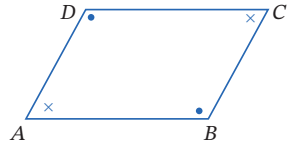
$\therefore ALMP$ is a parallelogram (one pair of opposite sides are parallel and equal)

EXERCISE 12.04 ANSWERS ON P. 672

Tests for quadrilaterals U F PS R C

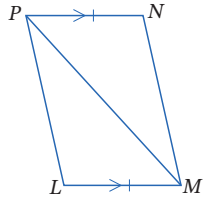
EXAMPLE 11

1 $ABCD$ is a quadrilateral in which opposite angles are equal. Prove that $ABCD$ must be a parallelogram. PS R C

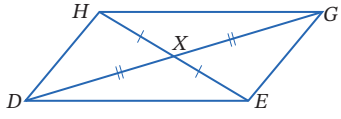


2 $LMNP$ is a quadrilateral in which $LM = NP$ and $LM \parallel NP$.

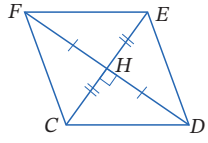
Prove that if a pair of opposite sides in a quadrilateral are equal and parallel, then the quadrilateral must be a parallelogram. PS R C



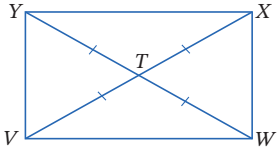
3 $DEGH$ is a quadrilateral whose diagonals DG and EH bisect each other. Prove that it must be a parallelogram. PS R C



4 $CDEF$ is a quadrilateral whose diagonals CE and DF bisect each other at right angles. Prove that $CDEF$ must be a rhombus. PS R C



5 $VWXY$ is a quadrilateral whose diagonals VX and WY are equal and bisect each other. Prove that it must be a rectangle. PS R C



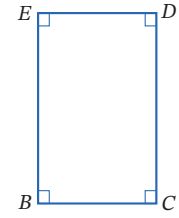
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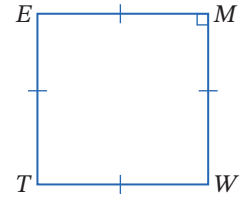
6 *BCDE* is a quadrilateral with all angles equal to 90°.

Prove that its opposite sides are parallel as well, and hence it must be a rectangle. PS R C



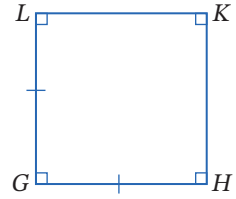
7 *TWME* is a quadrilateral with all sides equal and $\angle M = 90°$.

Prove that the other angles are 90° as well and hence it must be a square. PS R C



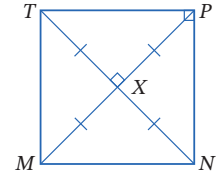
8 *GHKL* is a quadrilateral with all angles equal to 90° and $GH = GL$.

Prove that all sides are equal as well and hence it must be a square. PS R C



9 *MNPT* is a quadrilateral whose diagonals *MP* and *NT* are equal and bisect each other at right angles.

Prove that *MNPT* must be a square. PS R C

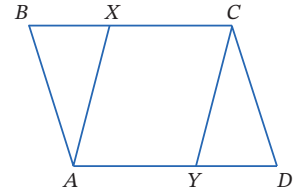


EXAMPLE 12

10 *ABCD* is a parallelogram and $BX = DY$. Prove that: R C

a $\triangle ABX \equiv \triangle CDY$

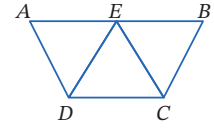
b *AXCY* is a parallelogram.



11 *AECD* is a rhombus and $AE = EB$. Prove that: R C

a $\triangle CBE \equiv \triangle DAE$

b *BCDE* is a parallelogram.



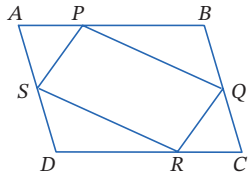
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12.04

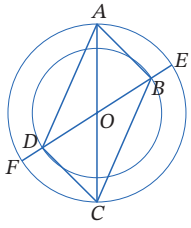
12 $ABCD$ is a parallelogram and $AP = AS = CQ = CR$.
Prove that: PS R C



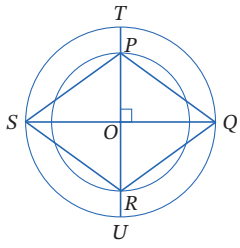
a $RQ = PS$ and $PQ = RS$

b $PQRS$ is a parallelogram

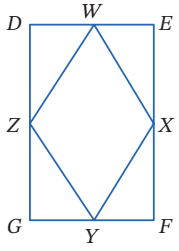
13 AC and DB are diameters of concentric circles with centre O.
Prove that $ABCD$ is a parallelogram. PS R C



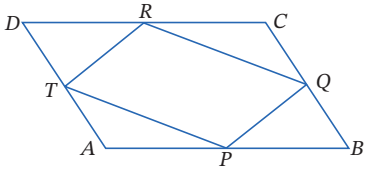
14 PR and SQ are diameters of concentric circles, centre O and $TU \perp SQ$. Prove that $PQRS$ is a rhombus. PS R C



15 $DEFG$ is a rectangle. W, X, Y and Z are the midpoints of the sides. Prove that $WXYZ$ is a rhombus. PS R C



16 $ABCD$ is a parallelogram. P, Q, R and T are the midpoints of the sides. Prove that $PQRT$ is a parallelogram. PS R C

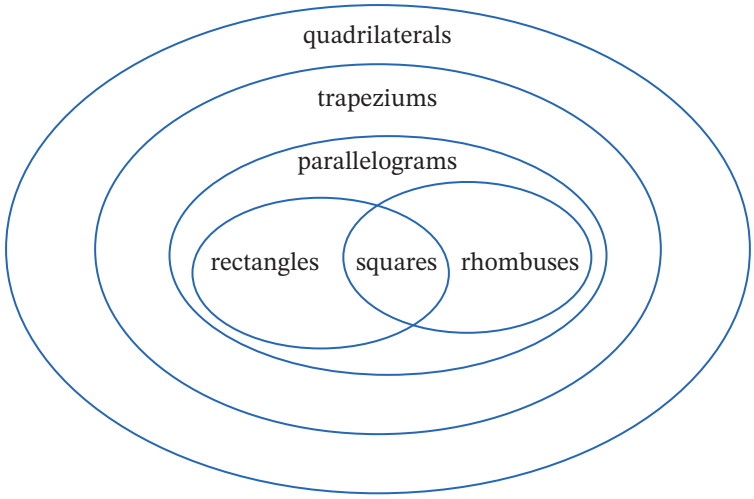


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Investigation

Is a square a rhombus?

Using the definitions of the special quadrilaterals, we see that a parallelogram can also be classified as a trapezium since it has at least one pair of opposite sides parallel. This means that trapeziums are **inclusive** of parallelograms. The set of trapeziums includes all parallelograms, or put another way, a parallelogram is a special type of trapezium. Similarly, parallelograms are inclusive of rectangles and rectangles are inclusive of squares. This can be represented by a Venn diagram:



1 Why is a rectangle a special type of parallelogram but a parallelogram is not always a rectangle? How can you use the Venn diagram to explain this?

2 Where would you put **kites** on this Venn diagram?

12.05 Proving properties of triangles and quadrilaterals

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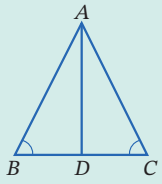
Worksheets
Quadrilaterals: True or false?

Proving properties of quadrilaterals

Properties of triangles and quadrilaterals can be proved using the congruence tests.

Example 13

$\triangle ABC$ is an isosceles triangle with $AB = AC$. D is the midpoint of BC.



a Which congruence test can be used to prove that $\triangle ABD \equiv \triangle ACD$?

b Explain why $\angle B = \angle C$.

c What geometrical result about isosceles triangles does this prove?

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Solution

a For $\triangle ABD$ and $\triangle ACD$:

$AB = AC$ (given)

AD is common.

$BD = CD$ (D is the midpoint of BC)

$\therefore$ The congruence test is SSS.

b $\angle B = \angle C$ because they are matching angles of congruent triangles.

c The angles opposite the equal sides of an isosceles triangle are equal.

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Presentation Geometric problems and proofs

12.05

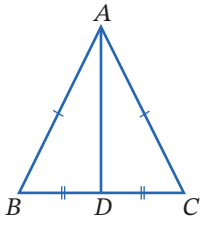
Video Proving properties of a rectangle

EXAMPLE 13

EXERCISE 12.05 ANSWERS ON P. 672

Proving properties of triangles and quadrilaterals U F R C

1 $\triangle ABC$ is an isosceles triangle, with $AB = AC$. D is the midpoint of BC. R C

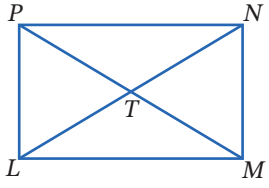


a Which congruence test can be used to prove that $\triangle ABD \equiv \triangle ACD$?

b Explain why $\angle ADB = \angle ADC$.

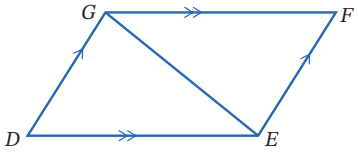
c Hence prove that $AD \perp BC$.

2 **a** If $LMNP$ is a rectangle, prove that $\triangle PNT \equiv \triangle MLT$.



b Prove that the diagonals of a rectangle bisect each other.

3 $DEFG$ is a parallelogram. R C

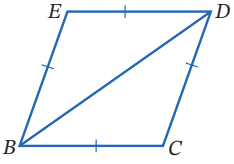


a Prove that $\triangle DEG \equiv \triangle FGE$.

b Draw the other diagonal DF and prove that $\triangle DGF \equiv \triangle FED$.

c Hence prove that the opposite angles of a parallelogram are equal.

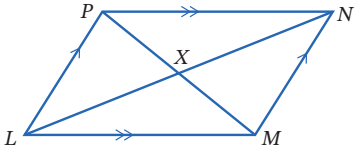
4 $BCDE$ is a rhombus, so all of its sides are equal and opposite sides are parallel. R C



a Prove that $\triangle BED \equiv \triangle BCD$.

b Hence prove that the diagonal BD bisects $\angle EBC$ and $\angle EDC$.

5 $LMNP$ is a parallelogram, so its opposite sides are parallel and equal. R C



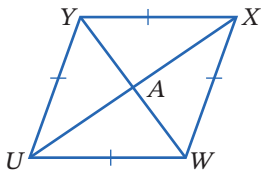
a Prove that $\triangle LXM \equiv \triangle NXP$.

b Hence prove that the diagonals of a parallelogram bisect each other.

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6 $UWXY$ is a rhombus. R C

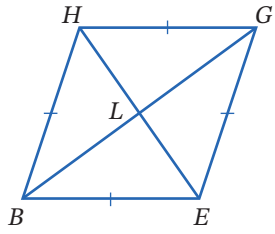


a Prove that $\triangle UAW \equiv \triangle XAW$.

b Prove that $\triangle UAW \equiv \triangle UAY$.

c Hence prove that the diagonals of a rhombus bisect each other at right angles.

7 $BEGH$ is a rhombus (a parallelogram with equal sides) whose diagonals BG and EH intersect at L. R C



a Prove that $\triangle BEL = \triangle GHL$.

b Prove that the diagonals of a rhombus bisect each other.

c $\triangle BEH$ is isosceles, so which equation is true?
Select **A**, **B**, **C** or **D**.

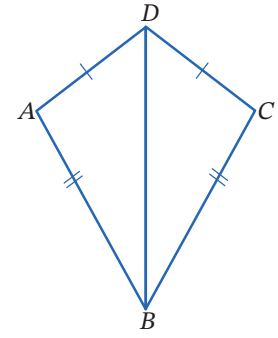
A $\angle BEH = \angle BHE$ **B** $\angle BLE = \angle BEH$

C $\angle HBE = \angle BEH$ **D** $\angle BEH = \angle EBH$

d Hence prove that $\triangle BEL \equiv \triangle BHL$.

e Hence prove that the diagonals of a rhombus cross at right angles.

8 $ABCD$ is a kite, so pairs of adjacent sides are equal. R C



a Prove that $\triangle ABD \equiv \triangle CBD$.

b Prove that $\angle A = \angle C$.

c Prove that diagonal DB bisects $\angle ADC$ and $\angle ABC$.

d Copy the diagram and draw the other diagonal AC, intersecting DB at point X.

e Prove that $\triangle DAX \equiv \triangle DCX$.

f Prove that diagonal DB bisects diagonal AC.

g Prove that $DB \perp AC$.

12.06 Formal geometrical proofs

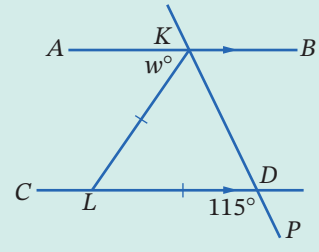
PATH

Puzzle
Geometrical proofs order activity

General geometrical results can be proved by writing a geometrical argument, where reasons are given at each step of the argument. This is called **deductive geometry.**

Example 14

In the diagram, $AB \parallel CD$, $KL = DL$ and $\angle LDP = 115°$.
Find the value of w, giving reasons.



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Solution

$\angle KDL = 180° - 115°$ (angles on a straight line)

$= 65°$

$\therefore \angle LKD = 65°$ (equal angles of isosceles $\triangle KLD$)

$\angle AKD = 115°$ (corresponding angles, $AB \parallel CD$)

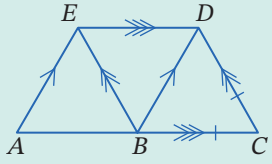
$\therefore w = 115° - 65°$

$= 50°$

Example 15

In the diagram, $AC \parallel ED$, $AE \parallel BD$, $BE \parallel CD$ and $CB = CD$.

Prove that $\triangle ABE$ is an isosceles triangle.



Solution

$\angle EBD = \angle CDB$ (alternate angles, $BE \parallel CD$)

and $\angle EBD = \angle AEB$ (alternate angles, $AE \parallel BD$)

$\therefore \angle CDB = \angle AEB$

But $\angle CDB = \angle CBD$ (equal angles of isosceles $\triangle CBD$)

and $\angle BAE = \angle CBD$ (corresponding angles, $AE \parallel BD$)

$\therefore \angle CDB = \angle BAE$

$\therefore \angle AEB = \angle BAE$

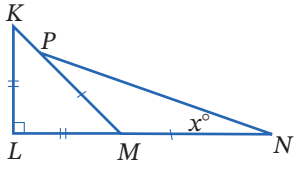
$\therefore \triangle ABE$ is an isosceles triangle (2 equal angles)

EXERCISE 12.06 ANSWERS ON P. 672

Formal geometrical proofs U F PS R C

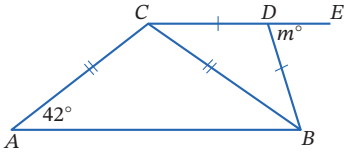
EXAMPLE 14

1 $KL = ML$ and $MN = MP$. Find x, giving reasons. PS R C

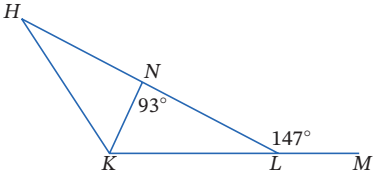


2 $CE \parallel AB$, $CD = BD$ and $AC = BC$.

Find m, giving reasons. PS R C



3 NK bisects $\angle HKL$. Find the size of $\angle NHK$, giving reasons. PS R C

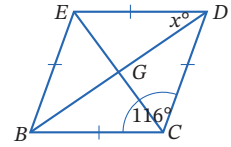


Foundation Standard Complex

PATH

4 $BCDE$ is a rhombus with the diagonals intersecting at G.

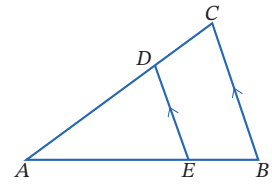
Find the value of x, giving reasons for each step. PS R C



EXAMPLE 15

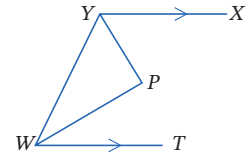
5 $\triangle ABC$ is an isosceles triangle where $AB = AC$ and $BC \parallel ED$.

Prove that $\triangle ADE$ is an isosceles triangle. PS R C



6 PY bisects $\angle XYW$, PW bisects $\angle TWY$ and $YX \parallel WT$.

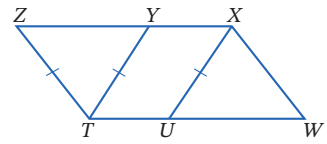
Prove that $\angle YPW = 90°$. PS R C



7 $TWXZ$ is a parallelogram and $TZ = TY = UX$. R C

a Prove that $\triangle TZY \equiv \triangle XWU$.

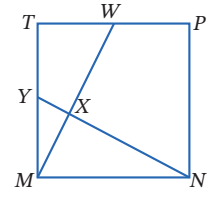
b Hence prove that $TUXY$ is a parallelogram.



8 $MNPT$ is a square. W and Y are the midpoints of sides TP and MT. R C

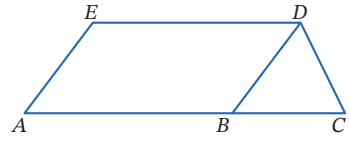
a Prove that $\triangle MNY \equiv \triangle MTW$.

b Prove that $MW \perp NY$.



9 $ABDE$ is a parallelogram and $BC = BD$.

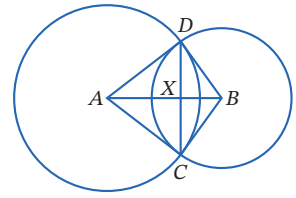
Prove that $\angle AED = 2\angle BCD$. PS R C



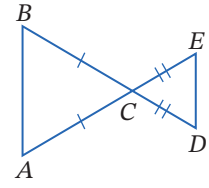
10 A and B are the centres of 2 circles that intersect at C and D. R C

a Prove that $\triangle ADB \equiv \triangle ACB$.

b Hence prove that $\triangle DXB \equiv \triangle CXB$ and $DX = CX$.



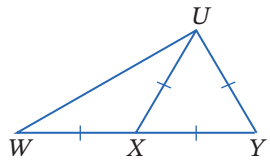
11 If $AC = BC$ and $DC = EC$, prove that $AB \parallel DE$. PS R C



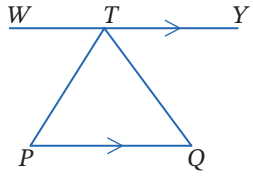
Foundation Standard Complex

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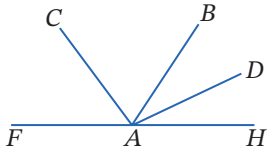
12 $\triangle UXY$ is an equilateral triangle and $WX = XU$. Prove that $\angle WUY$ is a right angle. PS R C



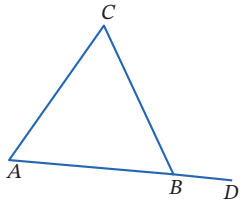
13 If $WY \parallel PQ$, prove that the angle sum of $\triangle PQT$ is 180°. PS R C



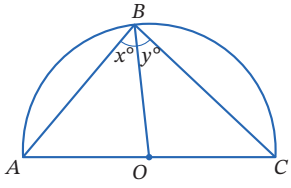
14 CA bisects $\angle FAB$ and DA bisects $\angle HAB$. Prove that $\angle CAD = 90°$. PS R C



15 Prove that the exterior angle of a triangle is equal to the sum of the interior opposite angles (that is, prove that $\angle CBD = \angle CAB + \angle BCA$). PS R C



16 AC is the diameter of a semicircle with centre O. B is a point on the semicircle. Let $\angle ABO = x°$ and $\angle CBO = y°$. Prove that $\angle ABC$ is a right angle. PS R C



PATH

12.06

Mental skills 12: Maths without calculators ANSWERS ON P. 672

Dividing a quantity in a given ratio

1 Study this example.

Divide \$5600 between Alice and Peter in the ratio 5 : 3.

Total number of parts = 5 + 3 = 8.

1 part = \$5600 ÷ 8 = \$700

Alice's share = 5 × \$700 = \$3500

Peter's share = 3 × \$700 = \$2100

Check: \$3500 + \$2100 = \$5600 (original amount)

Quiz
Mental skills 12

Foundation Standard Complex

2 Now divide each of these quantities in the given ratio.

- **a** Divide \$150 between Mark and Jenni in the ratio 2 : 1.
- **b** Divide \$2100 between Simon and Sunil in the ratio 4 : 3.
- **c** Divide \$720 between Lisa and Bree in the ratio 2 : 7.
- **d** Divide \$2000 between William and Adriana in the ratio 1 : 3.
- **e** Divide \$4500 between Anne and Pete in the ratio 3 : 2.
- **f** Divide \$3000 between Sharanya and Asam in the ratio 3 : 7.
- **g** Divide \$3600 between Cindy and Carmen in the ratio 5 : 1.
- **h** Divide \$1600 between Nancy and John in the ratio 3 : 5.
- **i** Divide \$990 between Carol and Louis in the ratio 5 : 4.
- **j** Divide \$4000 between Yvette and Andre in the ratio 1 : 4.
- **k** Divide \$4900 between Arden and Ivan in the ratio 3 : 4.
- **l** Divide \$3200 between Tan and Mai in the ratio 5 : 3.

12.07 Finding sides in similar figures

Video
The history of the golden ratio

Worksheets
A page of similar figures

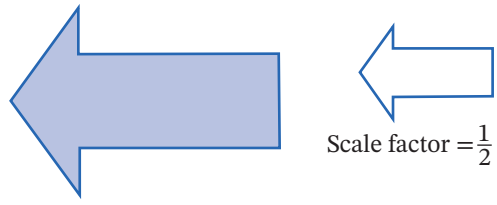
Enlargements and reductions

Enlarging a logo

Puzzle
Cartoon enlargement

Similar figures have the same shape, but are not necessarily the same size.

When a figure is enlarged or reduced, a **similar figure** is created.



Original **Image**

Scale factor

$$\text{Scale factor} = \frac{\text{image length}}{\text{original length}}$$

- If the scale factor is greater than 1, then the image is an enlargement.
- If the scale factor is between 0 and 1, then the image is a reduction.

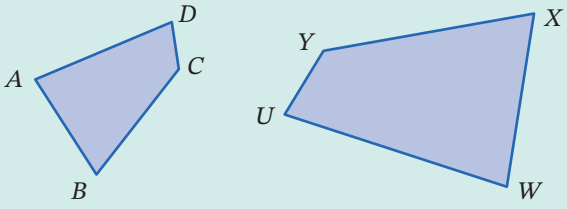
The symbol for 'is similar to' is '~'. When using this notation, we must make sure that the vertices (angles) of similar figures are written in matching order.

Properties of similar figures

- Matching angles are equal
- Matching sides are in the same ratio

Example 16

These 2 quadrilaterals $ABCD$ and $UWXY$ are similar.



Use the correct notation to write a similarity statement relating these 2 quadrilaterals.

Solution

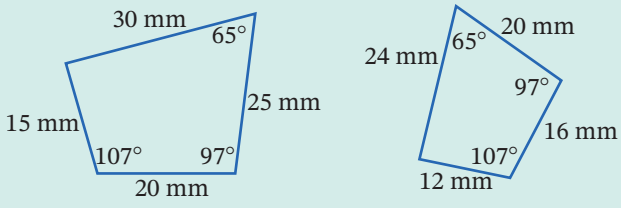
$\angle A$ matches with $\angle W$, $\angle B$ matches with $\angle X$, $\angle C$ matches with $\angle Y$, $\angle D$ matches with $\angle U$.

$\therefore ABCD \sim WXYU$ Matching order of vertices

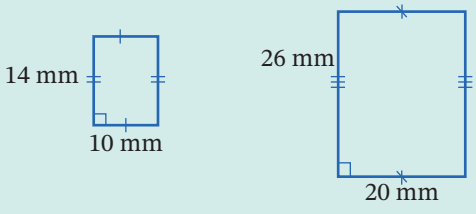
Example 17

Test whether each pair of figures are similar.

a



b



Solution

a For the 2 quadrilaterals, matching angles are equal and the ratios of matching sides are equal.

$\frac{20}{16}=\frac{5}{4}, \frac{25}{20}=\frac{5}{4}, \frac{30}{24}=\frac{5}{4}, \frac{15}{12}=\frac{5}{4}$

$\therefore$ The quadrilaterals are similar.

b For the 2 rectangles, matching angles are equal (90°), but the ratios of matching sides are not equal.

$\frac{10}{20}=\frac{1}{2}$ but $\frac{14}{26}=\frac{7}{13}$

$\therefore$ The rectangles are not similar.

Video
Finding an unknown side in similar figures

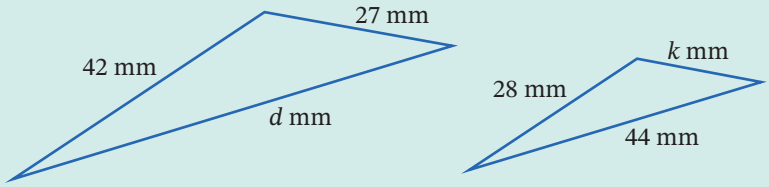
Skillsheet
Finding sides in similar triangles

Worksheet
Finding sides in similar figures

Puzzle
Similar triangles

Example 18

The 2 triangles are similar. Find the values of d and k.



Solution

$\frac{d}{44} = \frac{42}{28}$

$d = \frac{42}{28} \times 44$

$= 66$

$\frac{k}{27} = \frac{28}{42}$

$k = \frac{28}{42} \times 27$

$= 18$

Alternative method:

Scale factor $= \frac{28}{42} = \frac{2}{3}$

$d = 44 \div \frac{2}{3}$

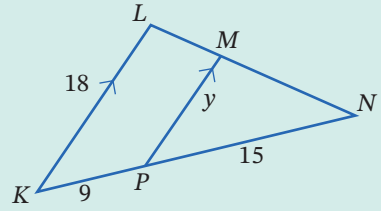
$= 66$

$k = 27 \times \frac{2}{3}$

$= 18$

Example 19

$\triangle KLN \sim \triangle PMN$. Find the value of y.



Solution

$\frac{MP}{LK} = \frac{PN}{KN}$ Ratios of matching sides are equal

$\frac{y}{18} = \frac{15}{24}$ $KN = 9 + 15 = 24$

$y = \frac{15}{24} \times 18$

$= 11\frac{1}{4}$

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EXERCISE 12.07 ANSWERS ON P. 672

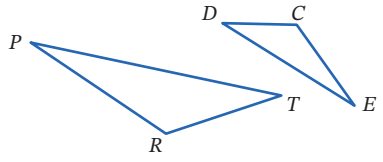
Finding sides in similar figures U F PS R C

1 For each pair of similar figures: use the correct notation ('~') to write a similarity statement relating them. R C

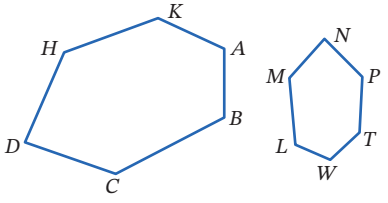
EXAMPLE 16

12.07

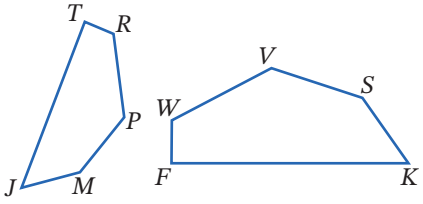
a



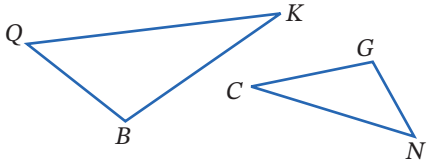
b



c



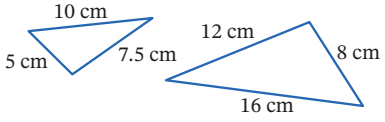
d



2 Test whether each pair of figures are similar. R C

EXAMPLE 17

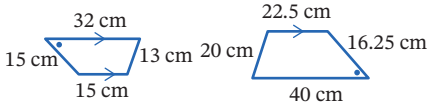
a



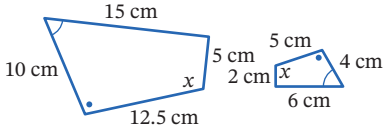
b



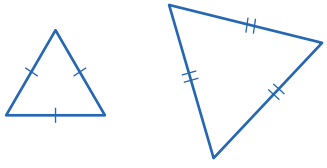
c



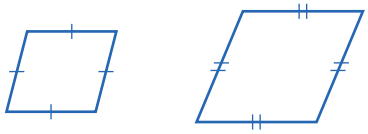
d



e



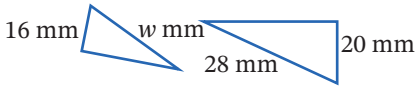
f



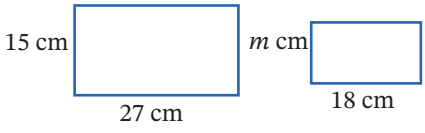
3 Find the value of each variable in each pair of similar figures. R

EXAMPLE 18

a

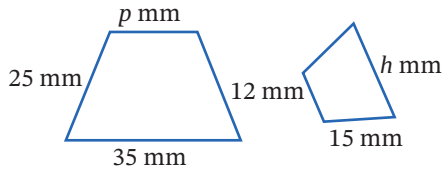


b

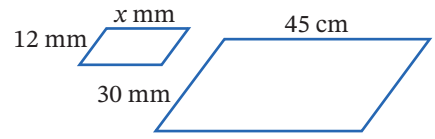


Foundation Standard Complex

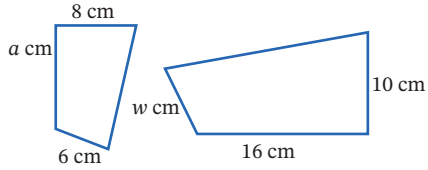
c



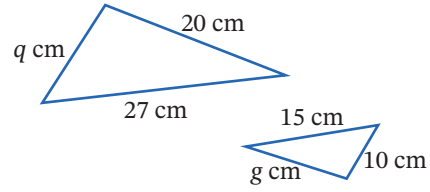
d



e

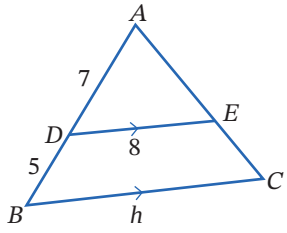


f

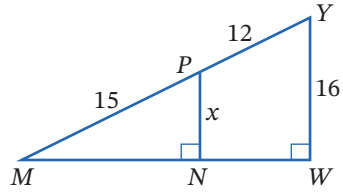


EXAMPLE 19

4 $\triangle ABC \sim \triangle ADE$.
Find the value of h. R



5 $\triangle MNP \sim \triangle MWY$.
Find the value of x. R

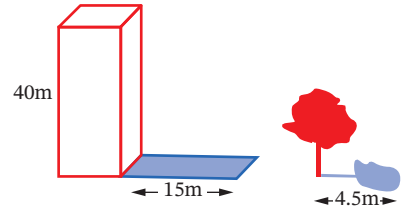


6 This photograph of the Sydney Harbour Bridge has been enlarged so that its length is 24 cm. If the dimensions of the original photo were 15 cm × 10 cm, what is the width of the enlargement? R



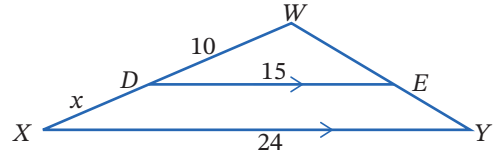
Taki O/Shutterstock.com

7 A building that is 40 m high casts a shadow 15 m long. At the same time, the shadow of a tree is 4.5 m long. What is the height of the tree? PS R



8 $\triangle WXY \sim \triangle WDE$. What is the value of x?
Select the correct answer **A**, **B**, **C** or **D**. R

A 4
B 6
C 8
D 10



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Tests for similar triangles

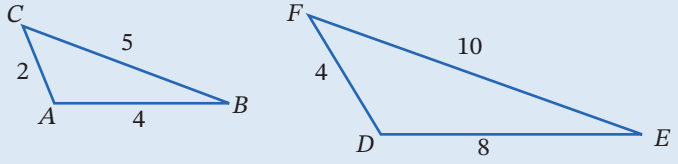
12.08

Similarity tests

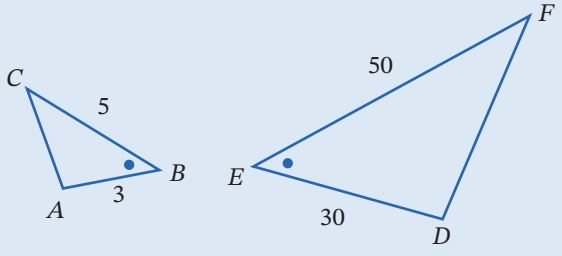
There are 4 tests for similar triangles.

Two triangles are similar if:

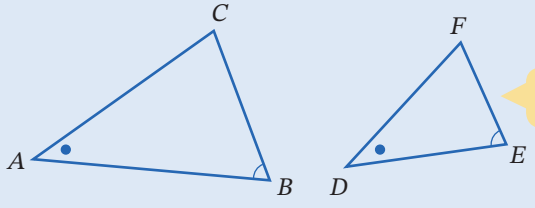
- the 3 sides of one triangle are proportional to the 3 sides of the other triangle (**'SSS'**)



- 2 sides of one triangle are proportional to 2 sides of the other triangle, and the **included angles** are equal (**'SAS'**)

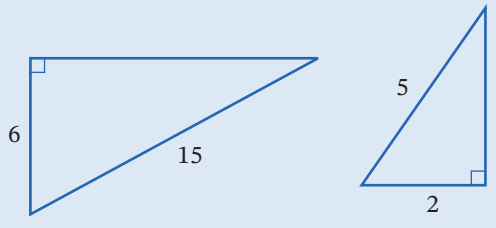


- 2 angles of one triangle are equal to 2 angles of the other triangle (**'AA'** or **'equiangular'**)



Equiangular means 'equal angles'

- they are right-angled and the hypotenuse and a second side of one triangle are proportional to the hypotenuse and a second side of the other triangle (**'RHS'**).



PATH

Worksheets
Congruent and similar triangles proofs

Congruence and similarity review

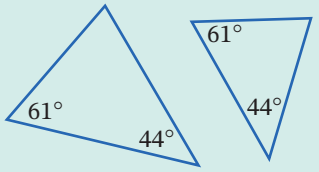
PATH

Video
Tests for similar triangles

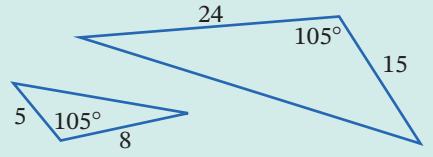
Example 20

Which test can be used to prove that each pair of triangles are similar?

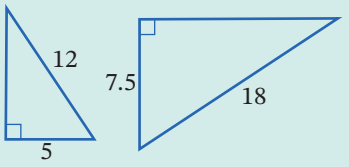
a



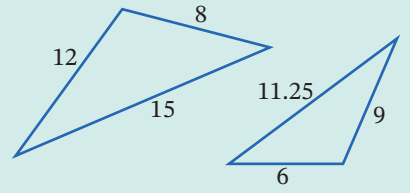
b



c



d



Solution

a 2 pairs of angles are equal, or equiangular ('AA').

b 2 pairs of matching sides are in the same ratio and the included angles in both triangles are equal ('SAS'). $\frac{15}{5}=\frac{24}{8}=3$

c Both have right angles, and the pairs of hypotenuses and second sides are in the same ratio ('RHS'). $\frac{7.5}{5}=\frac{18}{12}=\frac{3}{2}$

d All 3 pairs of matching sides are in the same ratio ('SSS'). $\frac{11.25}{15}=\frac{9}{12}=\frac{6}{8}=\frac{3}{4}$

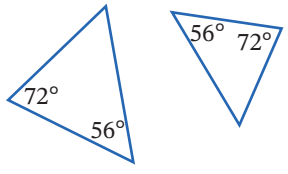
EXERCISE 12.08 ANSWERS ON P. 673

Tests for similar triangles U F R C

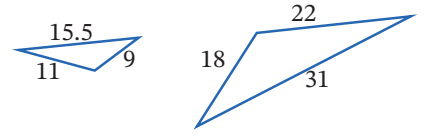
EXAMPLE 20

1 Which test can be used to prove that each pair of triangles are similar? R C

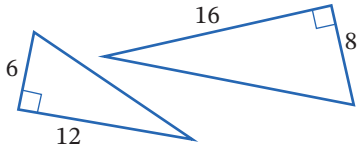
a



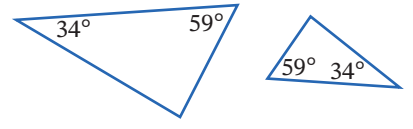
b



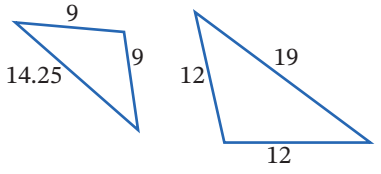
c



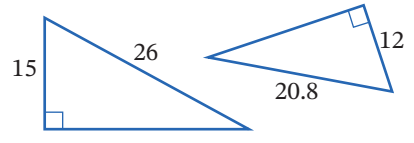
d



e



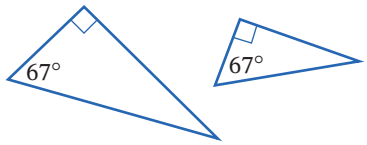
f



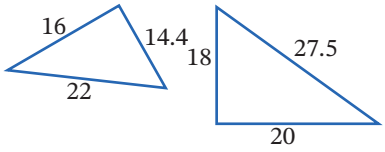
Foundation Standard Complex

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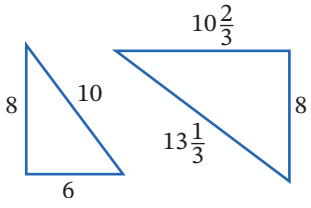
g



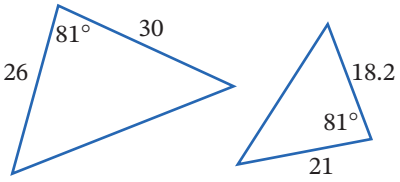
h



i

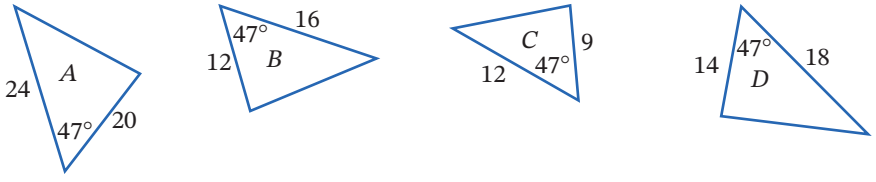


j

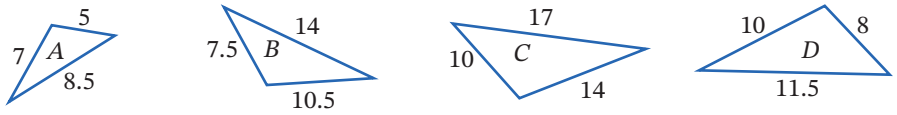


2 For each set of triangles, find the pair of similar triangles. R

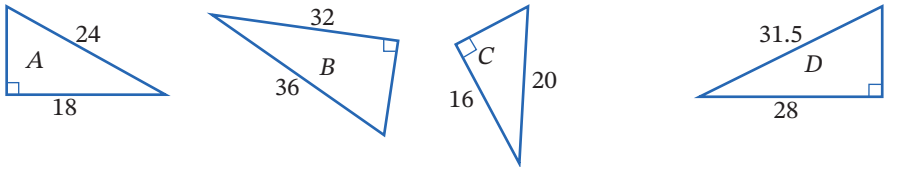
a



b

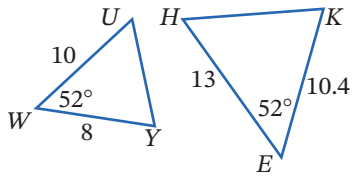


c

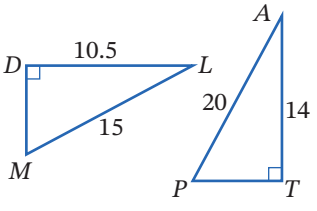


3 Use the correct notation to write a similarity statement relating each pair of similar triangles. R C

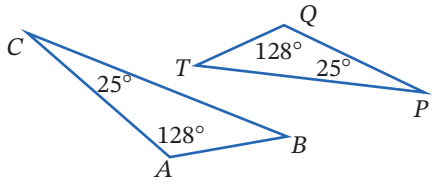
a



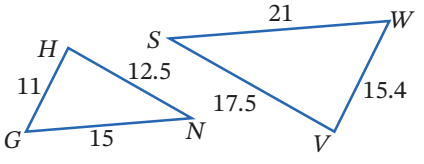
b



c



d



Foundation | Standard | Complex

12.09 Similar triangles proofs

PATH

Worksheets
Congruent and similar triangles proofs

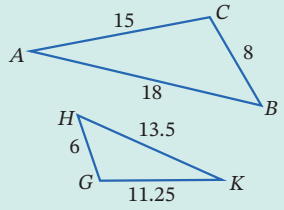
Complete the proofs

Video
Similar triangles proofs

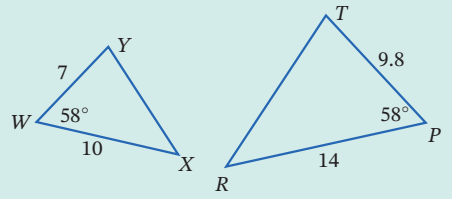
Example 21

Prove that each pair of triangles are similar.

a



b



Solution

a In $\triangle ABC$ and $\triangle KHG$:

$\frac{AB}{KH} = \frac{18}{13.5} = \frac{4}{3}$

$\frac{AC}{KG} = \frac{15}{11.25} = \frac{4}{3}$

$\frac{BC}{HG} = \frac{8}{6} = \frac{4}{3}$

$\therefore \frac{AB}{KH} = \frac{AC}{KG} = \frac{BC}{HG}$

$\therefore \triangle ABC \sim \triangle KHG$ (3 pairs of matching sides in proportion, or 'SSS')

b In $\triangle WXY$ and $\triangle PRT$:

$\frac{WX}{PR} = \frac{10}{14} = \frac{5}{7}$

$\frac{WY}{PT} = \frac{7}{9.8} = \frac{5}{7}$

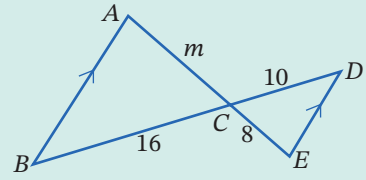
$\therefore \frac{WX}{PR} = \frac{WY}{PT}$

$\angle W = \angle P = 58°$

$\therefore \triangle WXY \sim \triangle PRT$ (2 pairs of matching sides in proportion and the included angles equal, or 'SAS')

Example 22

Prove that $\triangle ABC \sim \triangle EDC$ and hence find the value of m.



Solution

In $\triangle ABC$ and $\triangle EDC$:

$\angle A = \angle E$ (alternate angles, $AB \parallel DE$)

$\angle B = \angle D$ (alternate angles, $AB \parallel DE$)

$\therefore \triangle ABC \sim \triangle EDC$ (equiangular, or 'AA')

$\therefore \frac{m}{8} = \frac{16}{10}$ (matching sides in similar triangles)

$m = \frac{16}{10} \times 8$

$= 12.8$

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EXERCISE 12.09 ANSWERS ON P. 673

Similar triangles proofs U F R C

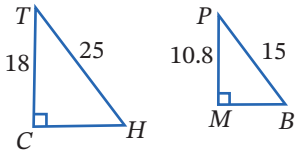
PATH

EXAMPLE 21

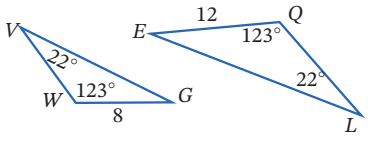
12.09

1 Prove that each pair of triangles are similar. R C

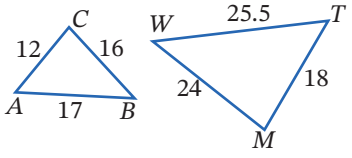
a



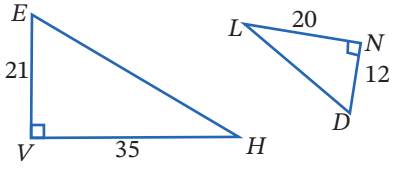
b



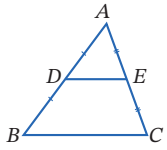
c



d

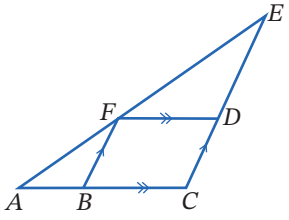


2 D and F are the midpoints of AB and AC.
Prove that $\triangle ADE \sim \triangle ABC$. R C

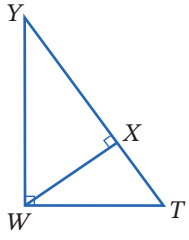


3 $AC \parallel FD$ and $BF \parallel CE$.
Prove that $\triangle ABF \sim \triangle FDE$. R C

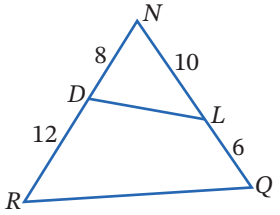
EXAMPLE 22



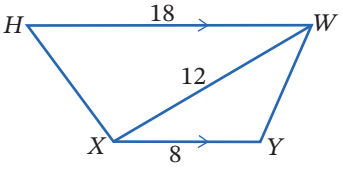
4 Prove that $\triangle WXY \sim \triangle TXW$. R C



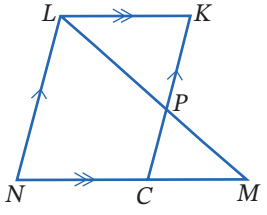
5 Prove that $\triangle NDL \sim \triangle NQR$. R C



6 $HW \parallel XY$. Prove that $\triangle XWH \sim \triangle XYW$. R C



7 $NCKL$ is a parallelogram.
Prove that $\triangle NML \sim \triangle KLP$. R C

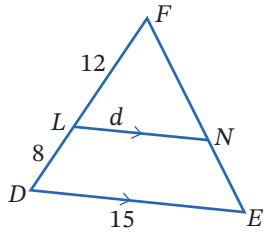


Foundation | Standard | Complex

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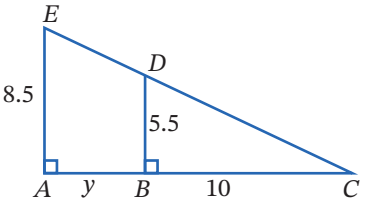
8 **a** Prove that $\triangle FLN \sim \triangle FDE$. R C

b Find the value of d.



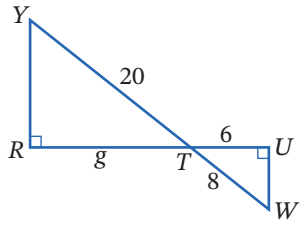
9 **a** Prove that $\triangle ACE \sim \triangle BCD$. R C

b Find the value of y.



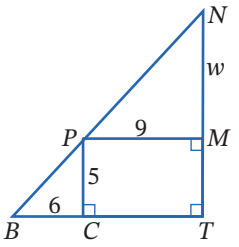
10 **a** Prove that $\triangle YRT \sim \triangle WUT$. R C

b Find the value of g.



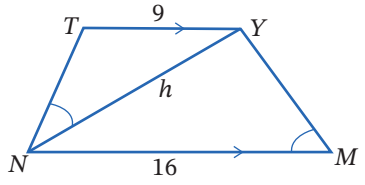
11 **a** Prove that $\triangle NMP \sim \triangle PCB$. R C

b Find the value of w.



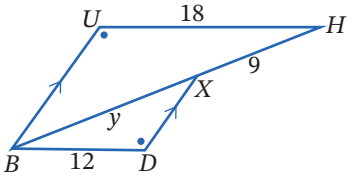
12 **a** Prove that $\triangle TYN \sim \triangle YNM$. R C

b Find the value of h.



13 **a** Prove that $\triangle BHU \sim \triangle XBD$. R C

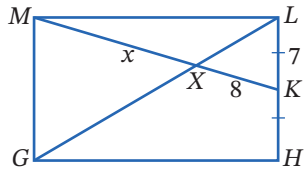
b Find the value of y.



14 $GHLM$ is a rectangle and K is the midpoint of HL. R C

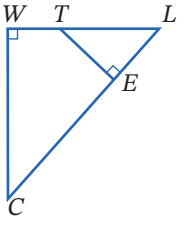
a Prove that $\triangle MXG \sim \triangle KXL$.

b Find the value of x.



15 **a** Prove that $\triangle CLW \sim \triangle LTE$. R C

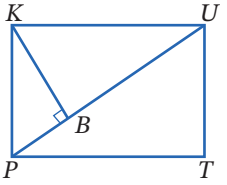
b If $WT = 5$ cm, $CE = 15$ cm and $EL = 6$ cm, find the length of TL.



16 $PTUK$ is a rectangle and $KB \perp PU$. R C

a Prove that $\triangle PTU \sim \triangle KBP$.

b If $BU = 21$ cm and $KP = 10$ cm, find the length of PB.



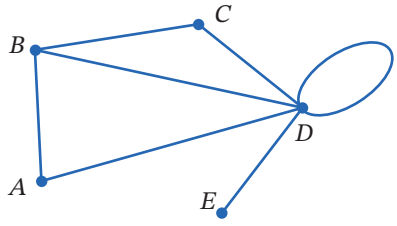
Foundation Standard Complex

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Networks 12.10

A **network** is a collection of objects connected to each other in some way or system. Examples of networks include bus, rail and air routes, computer systems and family or friend relationships.

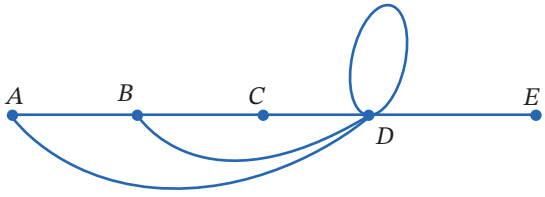
A **graph** is a diagram of a network where the objects are represented by points called **vertices** (or **nodes**), which are joined by lines called **edges**. The study of networks is called **graph theory**.



This graph or network diagram has 5 vertices, *A, B, C, D, E,* and 7 edges, *BC, BD, BA, AD, ED, CD, DD.*

A **loop** is an edge that connects a vertex to itself, such as *DD* in the diagram.

A network can be drawn in different ways as long as it shows the same connections. This diagram represents the same network as above.



PATH

Videos
Networks

The seven bridges of Konigsberg

Networks: Labyrinths and mazes

12.10

Degree of a vertex

The **degree** (or **order**) of a vertex is the number of edges connected to it, with loops counted as 2 edges. An **even vertex** has an even number of edges connected to it. An **odd vertex** has an odd number of edges.

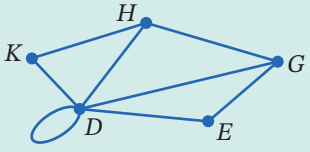
The **sum of the degrees** of all vertices in a network is equal to **twice the number of edges**, because each edge belongs to 2 vertices. This means that the sum of the degrees in a network is always even.

Example 23

a Find the degree of each vertex in this network.

b How many vertices are odd?

c Show that the sum of the degrees is equal to twice the number of edges.



Solution

a

Vertex	*D*	*E*	*G*	*H*	*K*
Degree	6	2	3	3	2

b There are 2 odd vertices.

c Number of edges = 8.

Sum of degrees of vertices = 6 + 2 + 3 + 3 + 2 = 16,

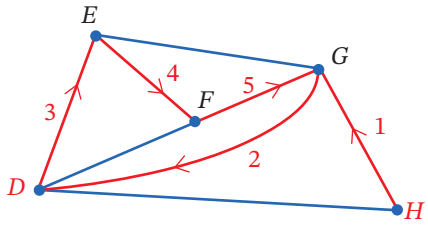
which is twice the number of edges (2 × 8).

PATH

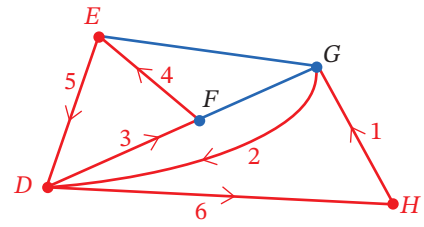
Walks, trails, paths and cycles

A **walk** is *any* route along the edges of a network, and may include repeated vertices or edges. Any walk that starts and ends at the same vertex is called a **closed walk**.

A **trail** is a walk with no repeated edges, such as shown in the network below. However, a vertex can be repeated, such as *G* in *HGDEFG*. A **circuit** is a **closed trail**, such as *HGDFEDH* shown below (starts and ends at *H*).



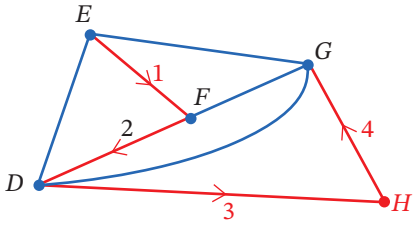
trail



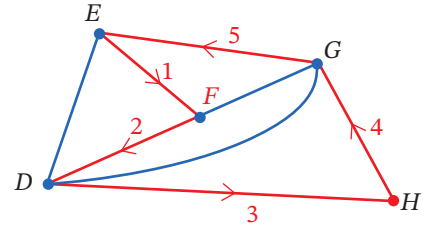
circuit

Think of a bushwalker walking on a trail or circuit and not repeating an edge.

A **path** is a walk with no repeated vertices (or edges), such as *EFDHG* shown in the network below. A **cycle** is a **closed path**, such as *EFDHGE* shown below (starts and ends at *E*, no repeated vertices).



path



cycle

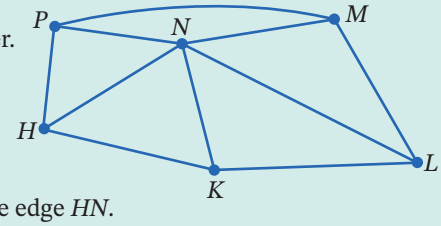
Think of a cyclist riding along a bicycle path or cycle and not revisiting any points.

Example 24

a State whether each walk is a trail, path or neither.

i *KLMPNHK* **ii** *PNMLNHK*

iii *HNMLNKHP* **iv** *LMNPHKNM*



b Write down a circuit that uses the edge *LN*.

c Write down a cycle the starts with *K* and uses the edge *HN*.

Solution

a **i** *KLMPNHK* is a path since no vertices (and edges) are repeated.

ii *PNMLNHK* is a trail as no edges are repeated but vertex *N* is repeated.

iii *HNMLNKHP* is a trail as no edges are repeated but the vertices *N* and *H* are repeated.

iv *LMNPHKNM* is neither as the edge *MN* is repeated.

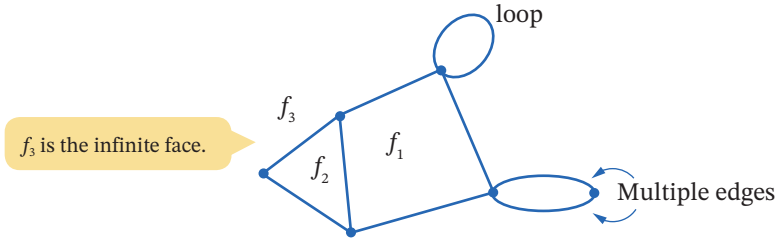
b *KLNMPHNK* is a circuit because it starts and ends at *K* with no repeated edges.

c *KHNPMLK* is a cycle because it has no repeated vertices.

Faces of a network

A **face** of a network is a region bounded by edges. Regions formed by loops or multiple edges between the same vertices are *not* faces, but the external (outside) region of a network is a face. The external face is called the **infinite face**.

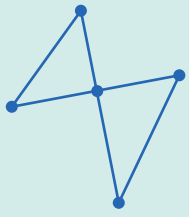
This network has 3 faces, f_1, f_2 and f_3.



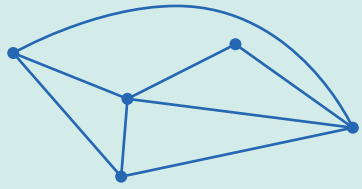
Example 25

Count the number of faces in each network.

a

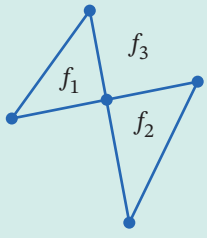


b



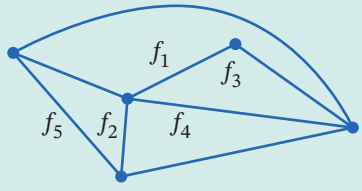
Solution

a



3 faces

b



5 faces

PATH

EXERCISE 12.10 ANSWERS ON P. 674

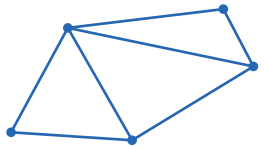
Networks U F R C

EXAMPLE 23

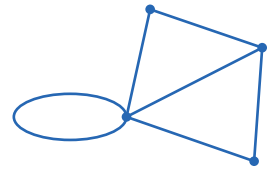
1 For each network, count the number of:

i vertices **ii** edges

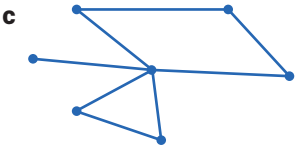
a



b



c

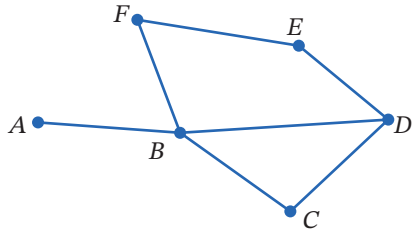


2 For each network: R C

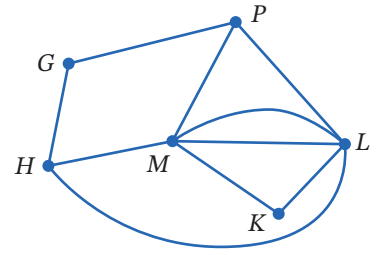
i find the degree of each vertex

ii show that the sum of the degrees is equal to twice the number of edges.

a



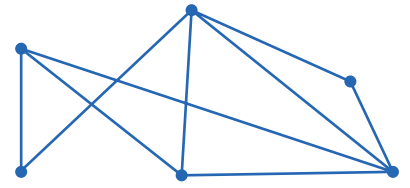
b



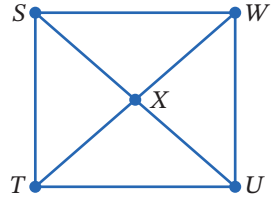
3 What is the sum of the degrees of all vertices in this graph? Select the correct answer **A**, **B**, **C** or **D**.

A 9 **B** 14

C 18 **D** 22

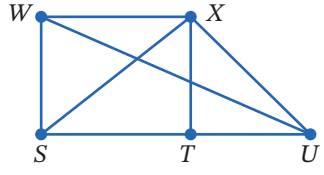


4 Which of these graphs shows the same information as the given network diagram?



Select **A**, **B**, **C** or **D**. R C

A



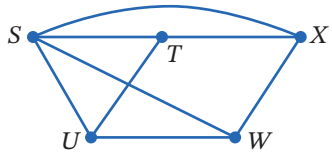
B



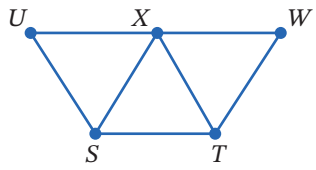
Foundation Standard Complex

9780170479660

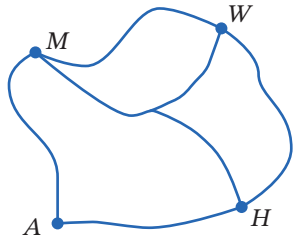
C



D

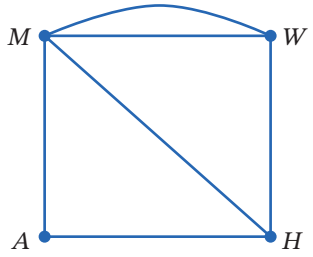


5 Which network diagram represents the map showing the roads that connect 4 towns?

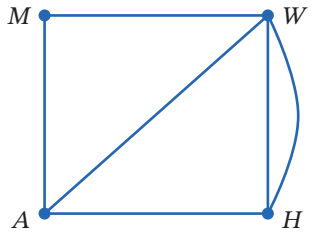


Select **A**, **B**, **C** or **D**. R C

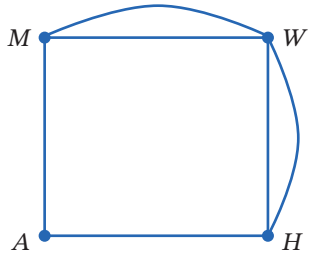
A



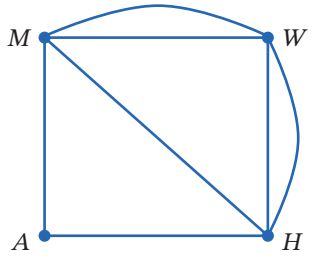
B



C



D



6 Anniek, Belinda, Emily, Mehek and Rima work at the same *J-Mart* store. R C

- Anniek is friends with Belinda, Emily and Rima
- Belinda is friends with Anniek and Emily
- Mehek is friends with Emily and Rima
- Rima is friends with Anniek, Mehek and Emily
- Emily is friends with Anniek, Belinda, Mehek and Rima

a Draw a graph to represent this network of friends.

b Who is not a friend with Anniek?

9780170479660

PATH

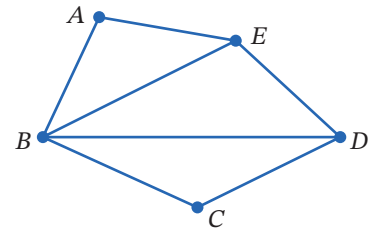
EXAMPLE 24

7 **a** State whether each walk is a trail or path. R C

i *AEBDCB* **ii** *EBDCBAE*

iii *EDCBAE* **iv** *BDCBAED*

v *DBEABCD* **vi** *DCBEA*



b Which of the walks in part **a** is:

i a circuit (a closed trail)?

ii a cycle (closed path)?

c Write down a circuit that contain the edges *BC* and *AE*.

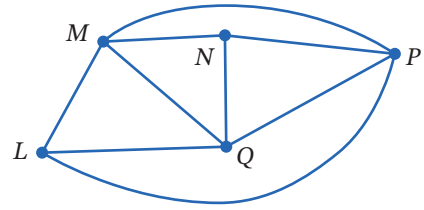
d Write down a cycle that contains the edge *CD*.

8 Which of the following walks is a circuit?

Select **A**, **B**, **C** or **D**. R C

A *MPQNMLPM* **B** *PMNPQLP*

C *QNPMLPN* **D** *LQPLMQP*

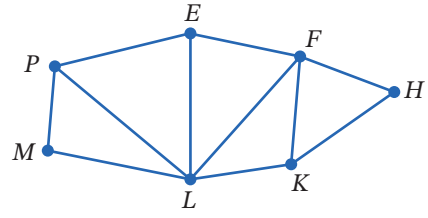


9 **a** Which of the following walks is a cycle?

Select **A**, **B**, **C** or **D**. R C

A *LKHFLEPM* **B** *FHEKHFEM*

C *MLKHFEKM* **D** *EFHKLPE*

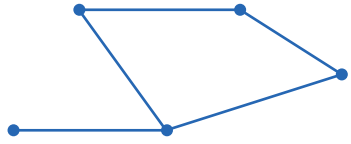


b Write down a cycle that begins at *K* and includes edge *LP*.

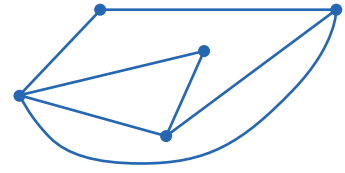
EXAMPLE 25

10 Count the number of faces in each network.

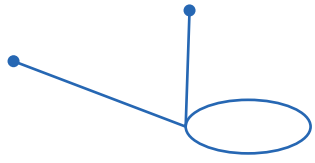
a



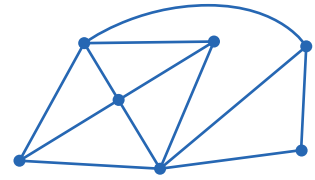
b



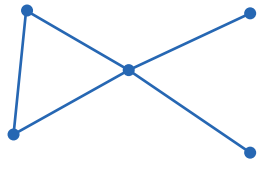
c



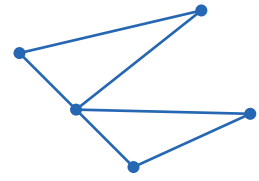
d



e



f

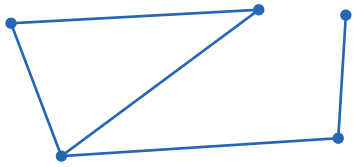


Foundation Standard Complex

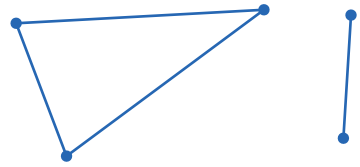
9780170479660

Planar graphs and Euler's formula

12.11



Connected



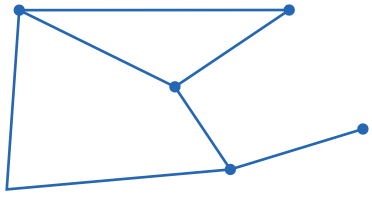
Not connected

PATH

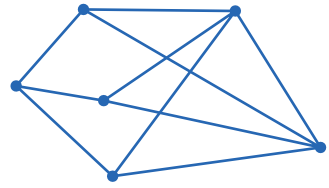
12.11

Videos
Euler's formula for planar graphs

A **planar graph** is a connected graph that can be drawn so that no edges cross over.

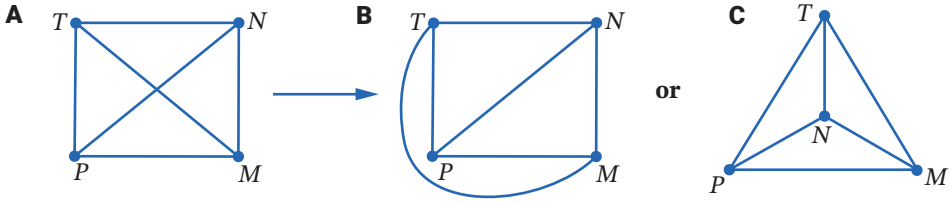


Planar graph



A non-planar graph

Some graphs may have edges crossing but they are still planar graphs because they can be *redrawn* without any of the edges crossing. For example, Graph **A** below is planar because it can be redrawn as **B** or **C**.

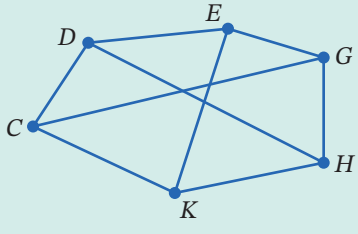


A **non-planar graph** has edges crossing and cannot be drawn so that the edges do not cross.

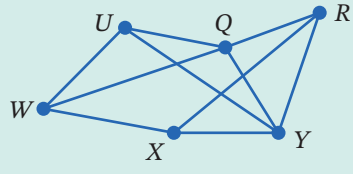
Example 26

State whether each network is a planar or non-planar graph.

a



b

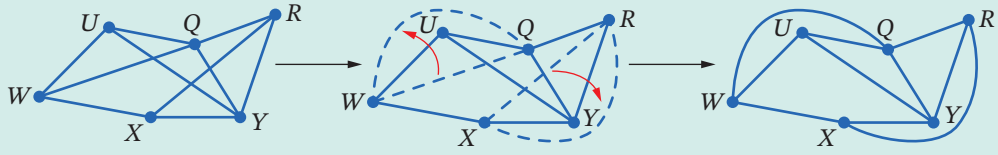


PATH

Solution

a It is impossible to move the edges so that no edges crossover so this is a non-planar graph.

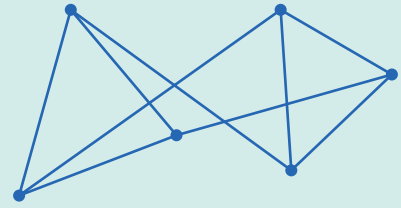
b This graph can be redrawn by moving XR and WQ as shown.



There are now no edges crossing so it's a planar graph.

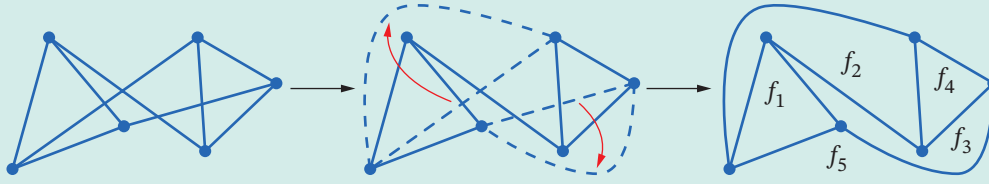
Example 27

Count the number of faces in this planar graph.



Solution

Redraw the graph so that no edges cross.



There are 5 faces.

Euler's formula

The Swiss mathematician Leonhard Euler (pronounced 'oiler') discovered a relationship between the vertices, edges and faces of a planar graph.

Euler's formula for planar graphs

For a graph to be planar,

$$v - e + f = 2$$

where, v = number of vertices, e = number of edges, f = number of faces.

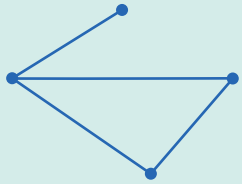
9780170479660

Example 28

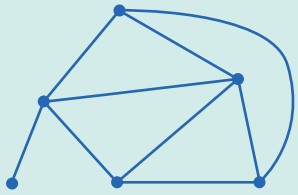
'Verify' means to show that something is true or correct.

Verify Euler's formula for each planar graph.

a



b



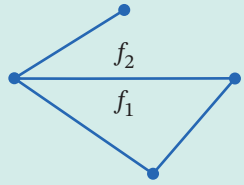
Solution

a $v = 4, e = 4$ and $f = 2$

$\therefore v - e + f = 4 - 4 + 2$

$= 2$

$\therefore$ Euler's formula is true.

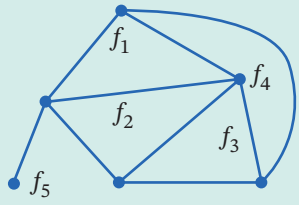


b $v = 6, e = 9$ and $f = 5$

$\therefore v - e + f = 6 - 9 + 5$

$= 2$

$\therefore$ Euler's formula is true.

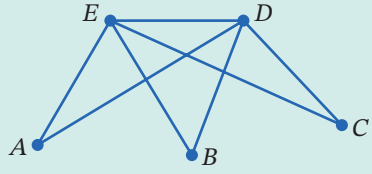


Example 29

Show that the graph is planar by:

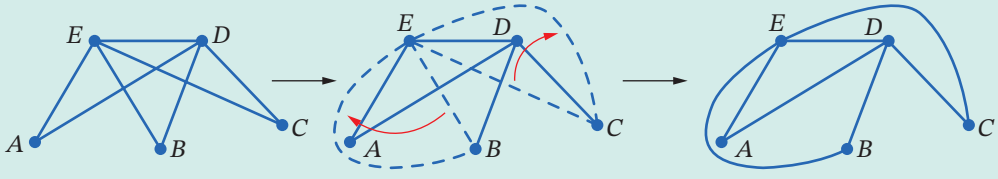
a redrawing the graph so no edges cross

b using Euler's formula



Solution

a Since the graph can be redrawn without any edges crossing, it is a planar graph.

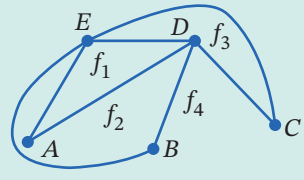


b $v = 5, e = 7$ and $f = 4$

$\therefore v - e + f = 5 - 7 + 4$

$= 2$

$\therefore$ the graph is planar



PATH

EXERCISE 12.11 ANSWERS ON P. 674

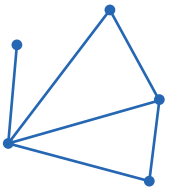
Planar graphs and Euler's formula U F R C

EXAMPLE 26

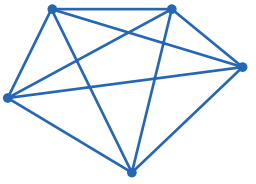
1 State whether each network is a planar or not planar graph.

Give a reason for your answer. C

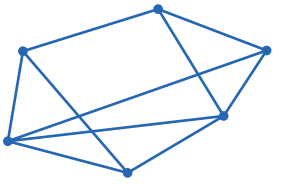
a



b

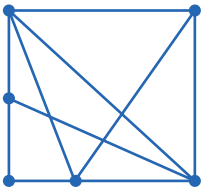


c

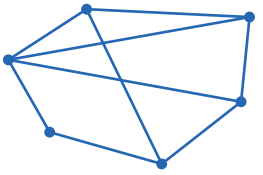


2 Redraw each graph to show that it is planar. C

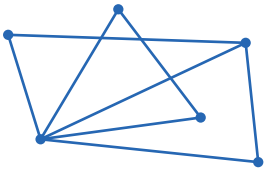
a



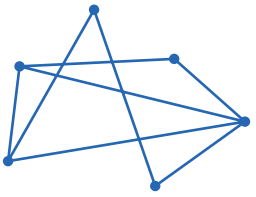
b



c



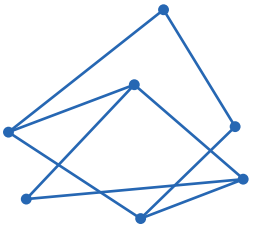
d



e



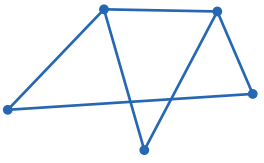
f



3 Redraw each planar graph to count its number of faces. R C

EXAMPLE 27

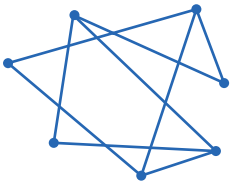
a



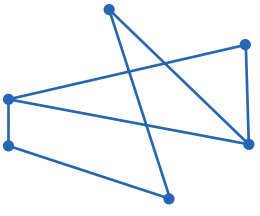
b



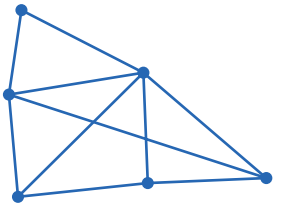
c



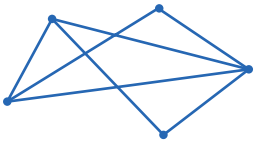
d



e



f



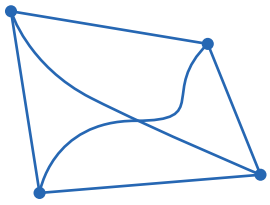
Foundation Standard Complex

9780170479660

PATH

12.11

4 How many faces does this graph have?



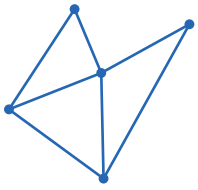
Select the correct answer **A**, **B**, **C** or **D**.

A 2 **B** 3

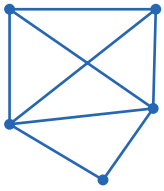
C 4 **D** 5

5 Which planar graph has 5 vertices and 5 faces? Select **A**, **B**, **C** or **D**.

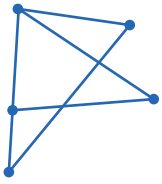
A



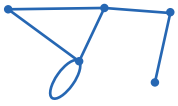
B



C



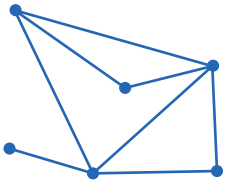
D



EXAMPLE 28

6 Verify Euler's formula for each planar graph. R C

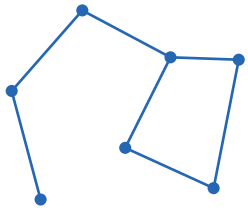
a



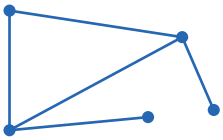
b



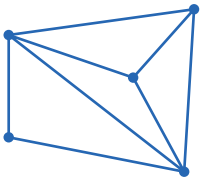
c



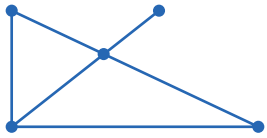
d



e



f

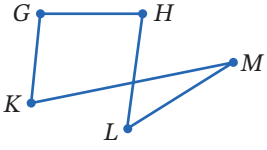


EXAMPLE 29

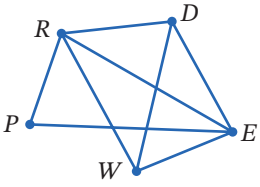
7 Show that each graph is planar by: R C

i redrawing the graph so no edges cross **ii** using Euler's formula

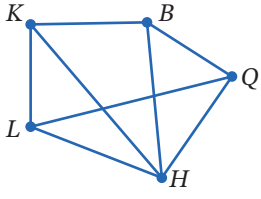
a



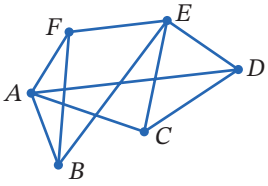
b



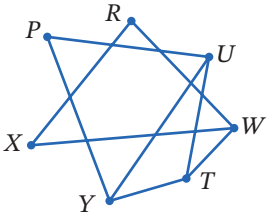
c



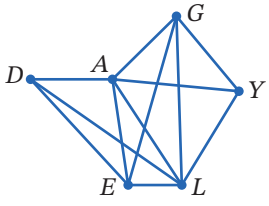
d



e



f



Foundation Standard Complex

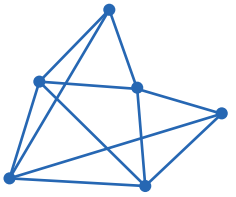
PATH

8 For which graph does Euler's formula NOT apply? Select **A**, **B**, **C** or **D**.

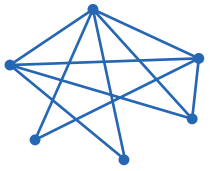
A



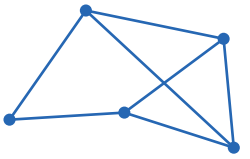
B



C



D



9 A planar graph has 7 vertices and 7 faces. What is its number of edges? Select **A**, **B**, **C** or **D**.

A 9 **B** 12 **C** 14 **D** 15

10 A planar graph has 10 faces and 13 edges. How many vertices has it? Select **A**, **B**, **C** or **D**.

A 3 **B** 5 **C** 8 **D** 11

11 A planar graph has 10 edges. Which of the following could be its number of vertices and faces? Select **A, B, C** or **D**.

A 9 vertices and 5 faces **B** 5 vertices and 5 faces

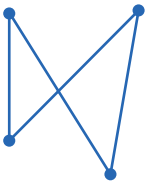
C 6 vertices and 7 faces **D** 8 vertices and 4 faces

Investigation

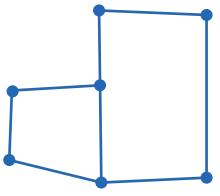
Eulerian trails and circuits

1 Copy each network, then trace over it with your pen or finger using each edge only once without lifting your pen or finger. Check your work with that of other students.

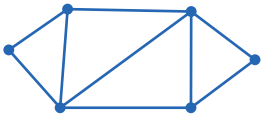
a



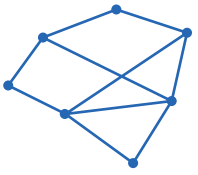
b



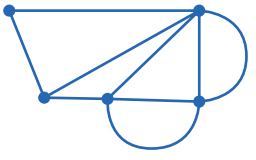
c



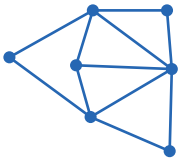
d



e



f



Foundation Standard Complex

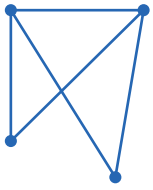
9780170479660

PATH

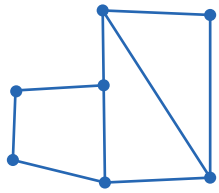
12.11

2 An extra edge has been added to each network in Question 1. Copy each network, then trace over it with your pen or finger using each edge only once. Which networks could you trace?

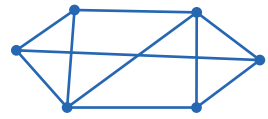
a



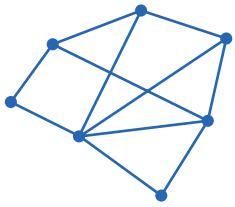
b



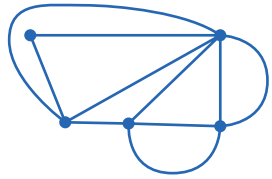
c



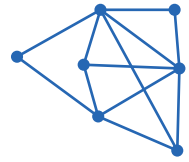
d



e



f



3 Copy and complete the table for each network in questions **1** and **2**.

Network	Number of odd vertices	Are all vertices even? (Y or N)	Is the network traceable? (Y or N)	Does trace begin and end at the same vertex? (Y or N)
1 a	0	Y	Y	Y
b				
c				
d				
e				
f				
2 a				
b				
c				
d				
e				
f				

4 Use the results in the table to copy and complete:

a A network can be traced if it is exactly _____ or _____ odd vertices.

b A network's trace starts and finishes at the same vertex if all vertices are _____.

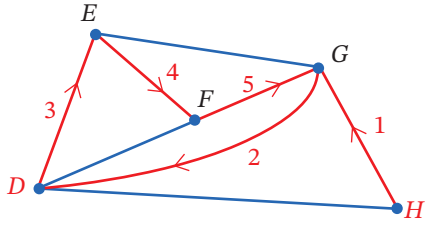
12.12 Eulerian trails and circuits

PATH

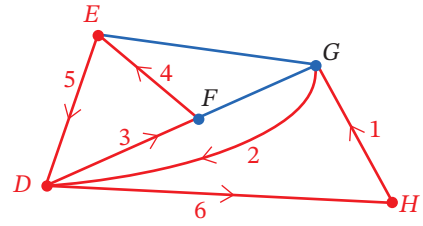
Video
Eulerian trails

Worksheet
Eulerian trails and circuits

Remember that a **trail** is a walk in a network with no repeated edges, while a **circuit** is a **closed trail**, starting and ending at the same vertex.

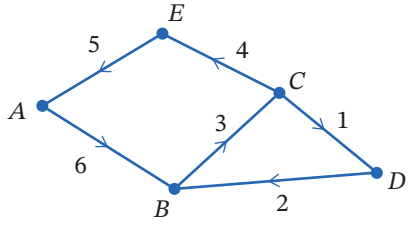


trail

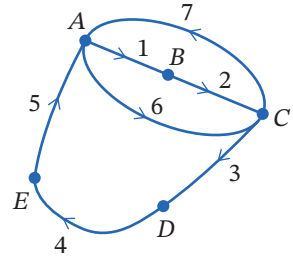


circuit

An **Eulerian trail** is a trail that uses *every* edge once, while an **Eulerian circuit** is a circuit that uses *every* edge once.



Eulerian trail



Eulerian circuit

Eulerian trails

An **Eulerian trail** uses every edge in the network only once.

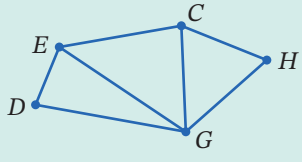
For an Eulerian trail to exist, the network must have exactly 2 odd vertices.

Eulerian trails start at one odd vertex and finish at the other odd vertex.

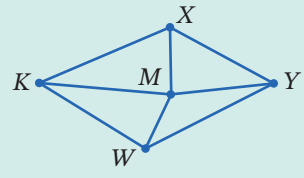
Example 30

Which network has an Eulerian trail?

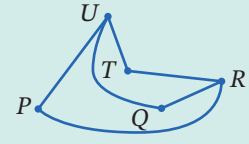
a



b



c



Solution

a This network has 2 odd vertices at C and E so it has an Eulerian trail, such as $EDGHCGEC$.

b This network has 4 odd vertices (degree 3) so it does not have an Eulerian trail.

c This network has 2 odd vertices at U and R (degree 3) so it has an Eulerian trail, such as $UPRQUTR$.

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Eulerian circuits

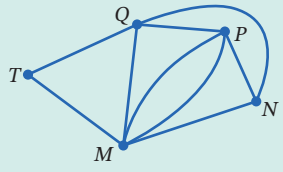
An **Eulerian circuit** uses every edge in the network only once and starts and ends at the same vertex.

For an Eulerian circuit to exist, all the vertices in the network must be even.

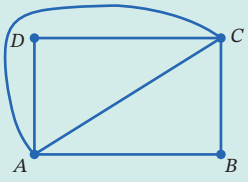
Example 31

Which network has an Eulerian circuit?

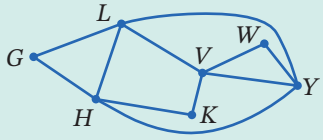
a



b



c



Solution

a Vertex M has degree 5 and vertex N has degree 3 so the network does not contain an Eulerian circuit.

b All vertices are even (B and D have degree 2, A and C have degree 4) so the network has an Eulerian circuit. Two possible Eulerian circuits are $BCADCAB$ and $ABCACDA$.

c All vertices are even (vertices G, K and W have degree 2, vertices H, L, V and Y have degree 4) so the network has an Eulerian circuit.

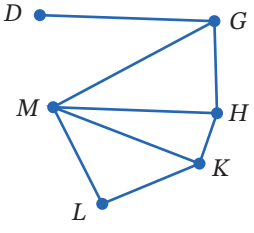
EXERCISE 12.12 ANSWERS ON P. 675

Eulerian trails and circuits U F R C

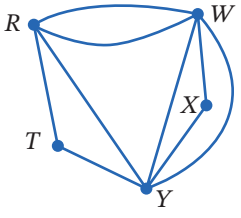
1 Which network has an Eulerian trail? Give reasons.

EXAMPLE 30

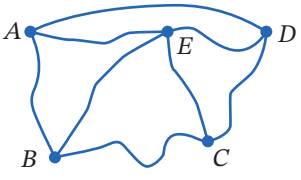
a



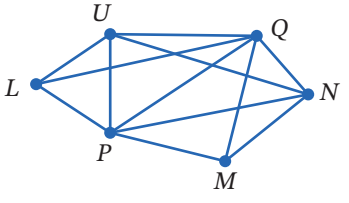
b



c



d



Foundation | Standard | Complex

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PATH

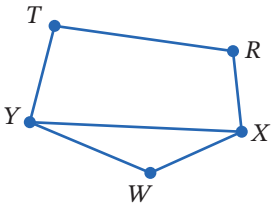
2 For each network: R C

i find the degree of each vertex

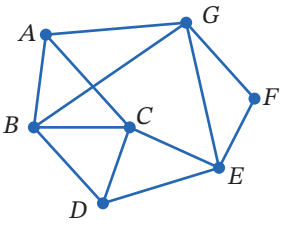
ii explain why the graph has an Eulerian trail

iii copy the graph and show a Eulerian trail on it

a



b

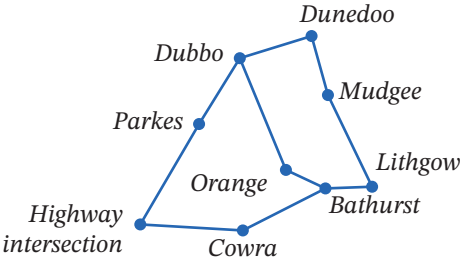


3 This graph shows the major roads linking cities in Central West NSW.

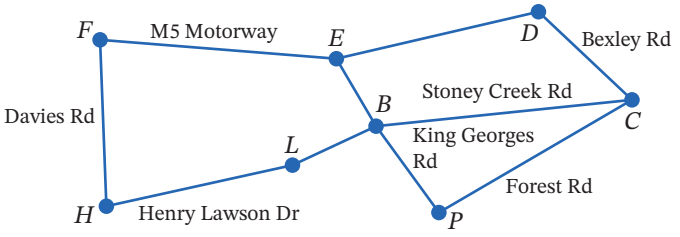
a Does this road network contain an Eulerian trail? Give reasons.

b Where will the trail start and finish?

c What is a possible Eulerian trail?



4 This graph shows a network of major roads in the St George area, south of Sydney. R C



a Explain why the graph has an Eulerian trail.

b If the trail starts at *E*, where will it finish?

c Copy the graph and show the trail.

5 Which vertices can be joined by an edge so that the network contains an Eulerian trail?

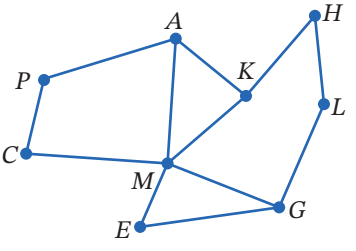
Select the correct answer **A**, **B**, **C** or **D**.

A *G* and *K*

B *M* and *P*

C *G* and *H*

D *K* and *L*



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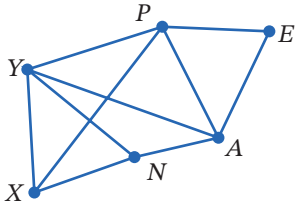
PATH

EXAMPLE 31

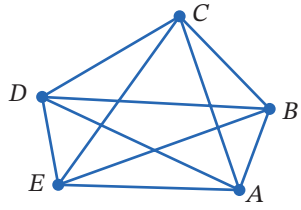
12.12

6 Test whether each network has an Eulerian circuit.

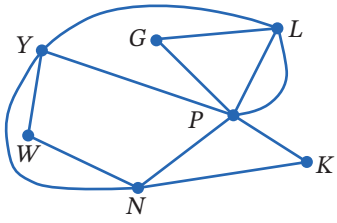
a



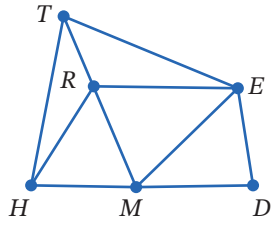
b



c



d

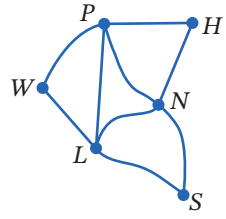


7 **a** Explain why this graph has an Eulerian circuit. R C

b Write down 2 Eulerian circuits.

c Copy the graph to draw the 2 circuits.

d What do you notice about your circuits?

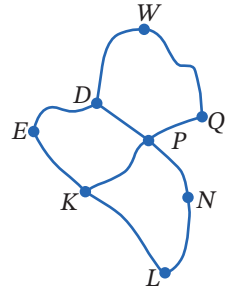


8 A road map for the delivery of milk to cafés and shops is shown by the network. The delivery starts at D and does not use the same road twice. R C

a Explain why the delivery cannot start and finish at D.

b A road is added to the network so that the delivery can start and finish at D. Which 2 vertices would the road join?

c Find 2 suitable routes for the milk delivery.



9 **a** This graph shows the roads connecting 4 towns.

Is it possible to start and finish at the same town after travelling along each road only once?

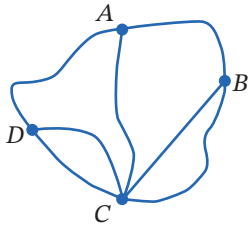
b A second road joining A to D is added.

Is it possible to travel along each road only once and

i start and finish at the same town?

ii start and finish at different towns?

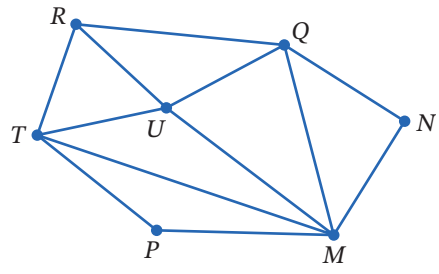
Give reasons. R C



Foundation | Standard | Complex

PATH

10 This tourist map shows 7 tourist destinations. Marinella and her family want to visit each destination, but use each road only once. R C



a From which destination should they start?

b At which destination will they finish their trip?

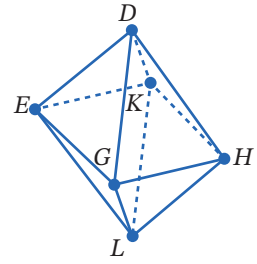
c How many destinations will Marinella and her family pass:

i only once? **ii** exactly twice?

11 An octahedron can be thought of as a 3D network.

a Show that this network has an Eulerian circuit.

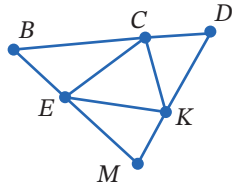
b Starting at *D*, describe a path that is an Eulerian circuit.



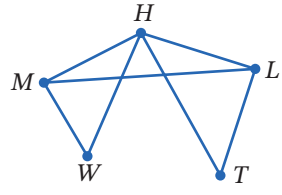
12 Determine which network is an Eulerian trail, an Eulerian circuit or neither.

Give reasons for your answers. R C

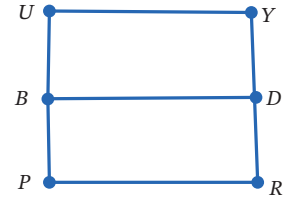
a



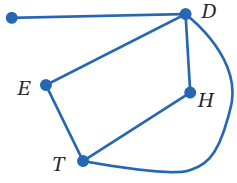
b



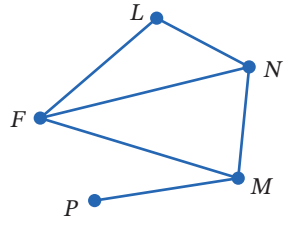
c



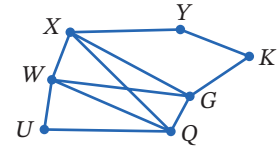
d



e



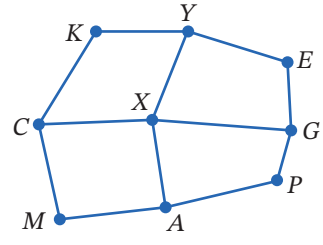
f



13 **a** Explain why this graph does not contain an Eulerian trail or circuit. R C

b How many vertices need to be joined for an Eulerian trail to exist in the graph?

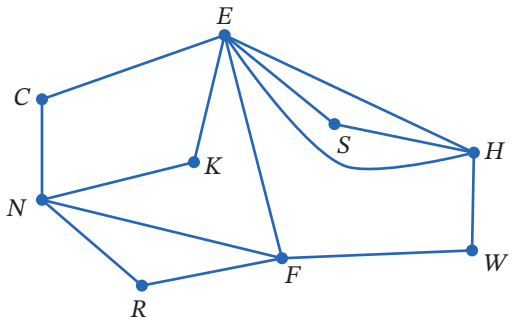
c Which vertices should be joined for an Eulerian circuit to exist in the graph?



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14 A marketing salesperson travels to stores in 8 different locations. In this network diagram, the vertices represent the locations and the edges represent the roads. R C



a Explain why the salesperson can travel from C, visit every location using each road only once and then return to location C.

b Describe the route the salesperson could take.

PATH

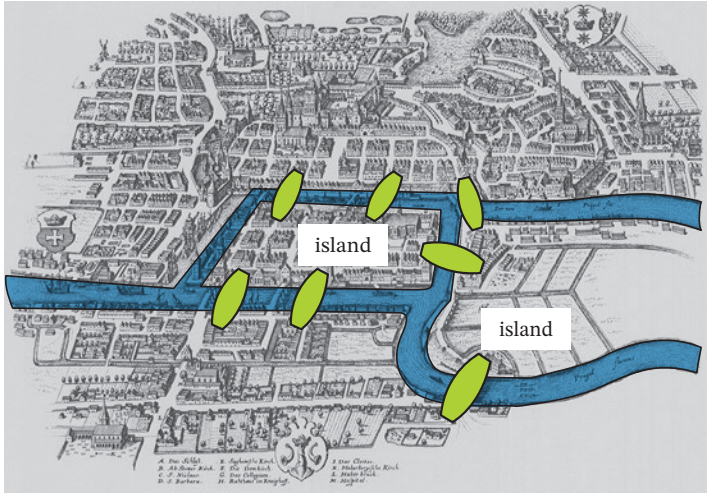
12.12

Did you know?

The 7 bridges of Königsberg

Video
The seven bridges of Konigsberg

In the Prussian town of Königsberg (which is now Kaliningrad, Russia), there were 7 bridges connecting 2 islands and the banks of the Pregel River. There was a tradition where people would walk through the town on a Sunday and it became a challenge to complete a walk crossing each bridge only once. No person was able to do it but no one could prove that it was impossible.



Album/Alamy Stock Photo

In 1736, the Swiss mathematician Leonard Euler solved the problem by reducing the map to a network. He found that it was impossible to do the walk. Eventually another bridge was built. That is why a walk that crosses every bridge exactly once is called an **Eulerian trail**. Euler developed a new branch of mathematics called **graph theory**, the geometry of networks.

1 **Draw a network to represent the map and explain why the original walk was impossible.**

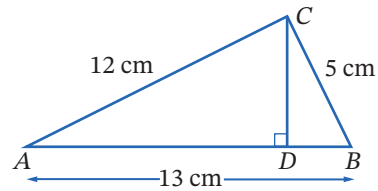
2 **An eighth bridge was built. Where would the bridge have been placed so that the network could be traversed?**

Foundation Standard Complex

Power plus ANSWERS ON P. 676

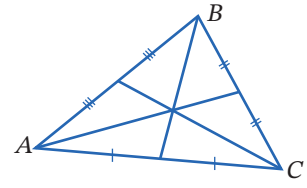
1 **a** Prove that the 3 triangles *ABC*, *ACD* and *CBD* are similar.

b Find the length of *CD*.



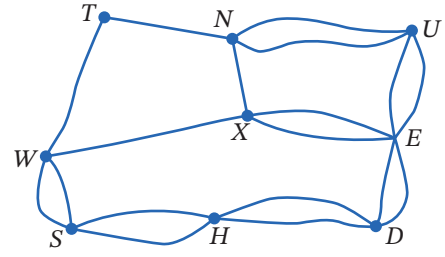
2 *ABCD* is a parallelogram. *HG* is any interval joining parallel sides *AB* and *DC* and passing through the midpoint of the diagonal *BD*. Prove that the interval through the midpoint of a diagonal in a parallelogram divides opposite sides equally. (That is, prove that $DG = BH$.)

3 The median is a line joining a vertex of a triangle to the midpoint of the opposite side. Prove that the medians of a triangle are concurrent (that is, they meet at one intersection point).



4 Leilani is surveying homes on family size.

Some streets have homes on both sides (shown by double edges on the network diagram) and other streets only have homes on one side only (shown by single edges).



a Explain why an Eulerian circuit can be found on this network.

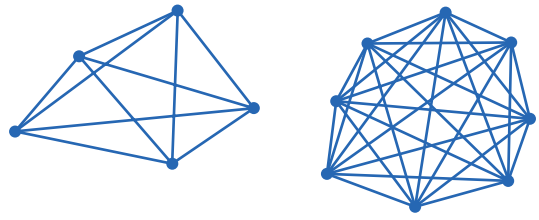
b Write down a circuit that begins at:

i *N* **ii** *H* **iii** *X*

c For each of your circuits from part **b**, write down which vertices occur:

i once in the circuit **ii** 3 times in the circuit

5 In a polygon, each vertex is joined to all other vertices as shown (all diagonals are shown).



What type of polygon drawn this way will always have an Eulerian circuit?

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CHAPTER 12 REVIEW

Language of maths

AAS	congruence test	congruent (≡)	convex polygon
degree	edge	equiangular	Eulerian
exterior angle	face	included angle	planar graph
reflection	regular polygon	rotation	RHS
SAS	scale factor	similar (~)	similarity test
SSS	transformation	translation	vertex (node)

1 What is a **convex polygon**?

2 Explain the difference between the **interior** and **exterior** angles of a polygon.

3 What is the symbol and meaning of 'is similar to'?

4 What is the meaning of the 'A' in the SAS test for congruent triangles?

5 What does **equiangular** mean in the similarity tests?

6 Explain the difference between an **Eulerian trail** and an **Eulerian circuit**.

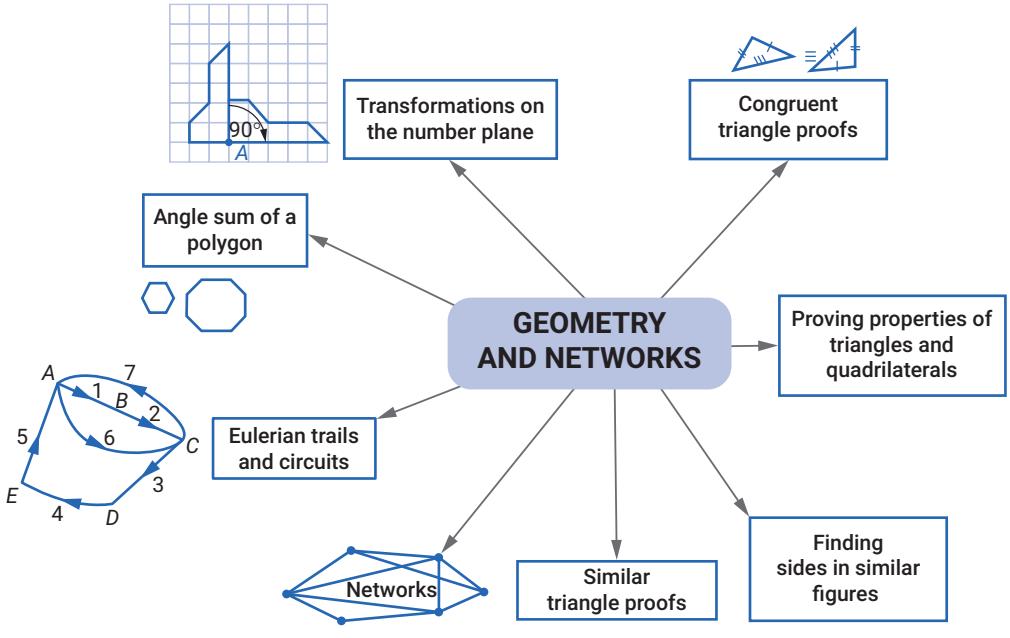
Quiz
Language of maths 12

Puzzles
Geometry and networks crossword
Geometry crossword (Advanced)

Topic summary

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

Worksheet
Mind map: Geometry and networks (Advanced)



Quiz
Test yourself 12

TEST YOURSELF 12

ANSWERS ON P. 676

PATH

12.01

1 Find the angle sum of a polygon with:

a 15 sides **b** 24 sides **c** 12 sides **d** 48 sides

2 Find the size of one angle in a regular 15-sided polygon.

3 The angle sum of a polygon is 6120°. How many sides does the polygon have?

4 Find the number of sides in a regular polygon if each exterior angle is:

a 10° **b** 24° **c** 45° **d** 15°

12.02

5 **a** Copy the quadrilateral *PLMN* onto grid paper and then draw its image after:

i reflecting it across the *y*-axis

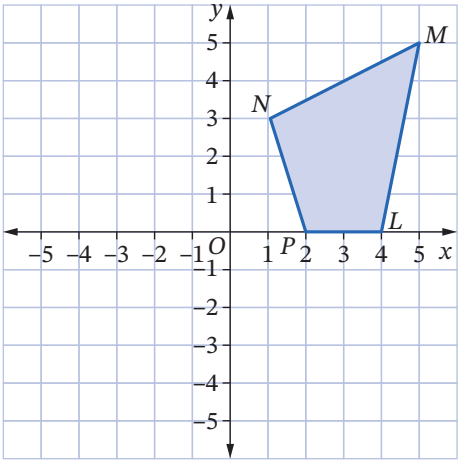
ii reflecting it across the *x*-axis

iii rotating it anticlockwise about *O* through 90°

iv translating it 5 units to the left and 4 units down

b For each of the above transformations, compare the coordinates of each vertex of the original to its matching vertex in its image.

Worksheet
5 mm grid paper



12.02

6 Find the new position of the point $K(-5, 3)$ if it is:

a reflected in the *y*-axis **b** reflected in the *x*-axis

c rotated clockwise about *O* through

i 90° **ii** 180° **iii** 270°

d translated 8 units to the right and 5 units down

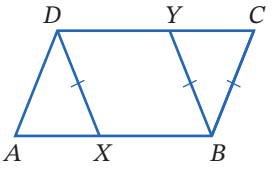
e translated 3 units to the right, 2 units up and then rotated 180° about *O*.

12.03, 12.04

7 *ABCD* is a parallelogram and $BC = BY = DX$.

a Prove that $\triangle DAX \equiv \triangle BCY$.

b Hence, prove that *BXDY* is a parallelogram.



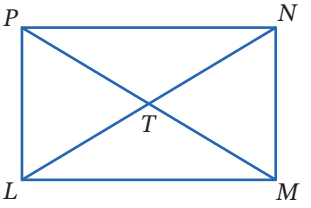
8 *PNML* is a rectangle.

12.05

a Prove that $\triangle PML \equiv \triangle NLM$.

b Hence explain why $PM = NL$.

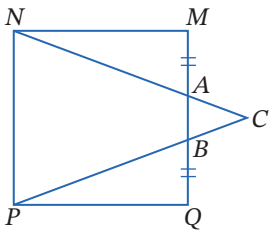
c What geometrical result about rectangles does this prove?



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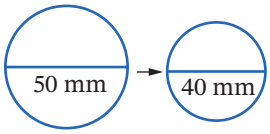
9 $PNMQ$ is a square and $AM = BQ$. Prove that $\triangle NPC$ is isosceles.



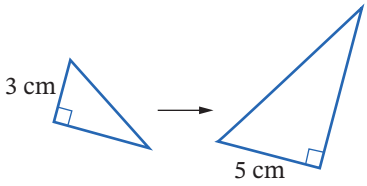
PATH 12.06

10 Calculate the scale factor between each pair of similar figures. 12.07

a

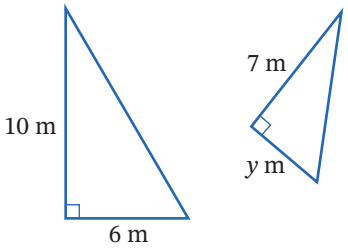


b

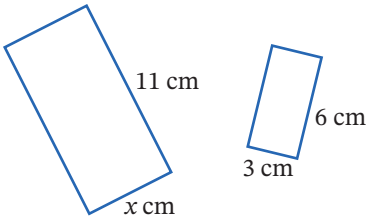


11 Find the value of the variable in each pair of similar figures. 12.07

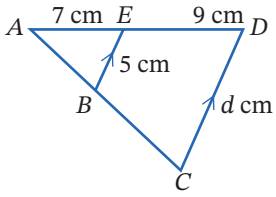
a



b



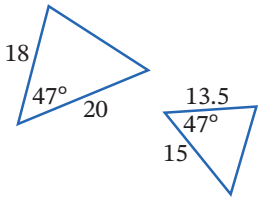
12 If $\triangle ABE \sim \triangle ACD$, find the value of d.



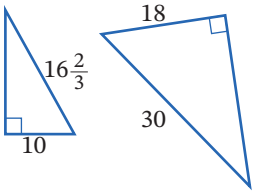
PATH 12.07

13 Which test can be used to prove that each pair of triangles are similar? 12.08

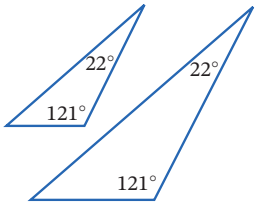
a



b



c

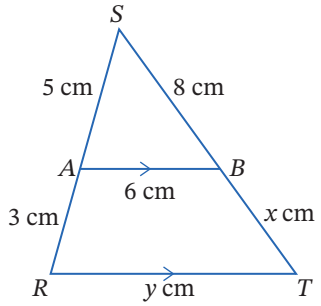


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PATH

12.09

14 Prove that $\triangle ASB \sim \triangle RST$ and hence find the values of x and y.



12.09

15 Use similar triangles to prove that the interval joining the midpoint of 2 sides in a triangle is parallel to the third side and is half its length.

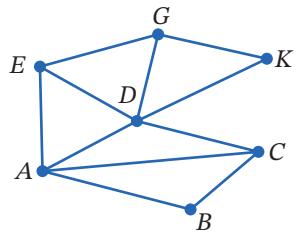
12.10

16 **a** State whether each walk is a trail or a path.

i *EABCDKG* **ii** *CADKGEABC*

iii *EDKGDAB* **iv** *DCBAEGKD*

b Which of the walks in part **a** is a **circuit**?



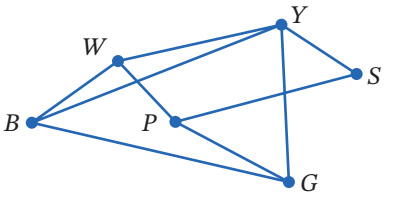
12.10

17 How many faces does this graph have?

Select **A**, **B**, **C** or **D**.

A 5 **B** 6

C 7 **D** 8

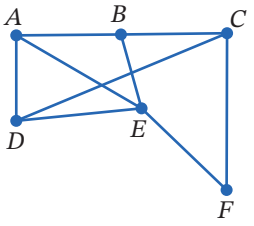


12.11

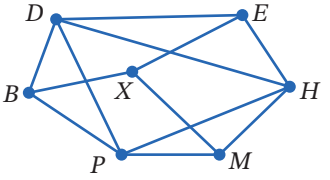
18 Show that the graph is planar by:

i redrawing it so that no edges cross **ii** using Euler's formula

a



b



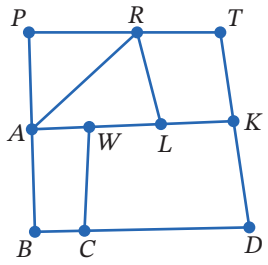
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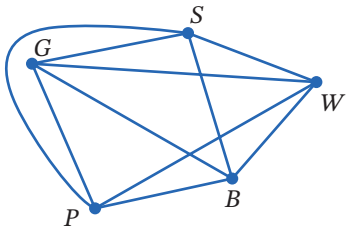
19 Determine whether each network has an Eulerian trail, an Eulerian circuit or neither. Give a reason for your answer.

12.12

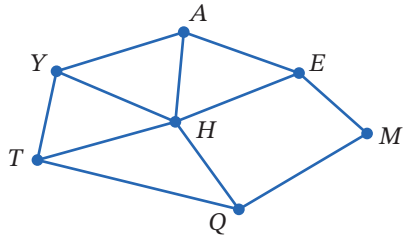
a



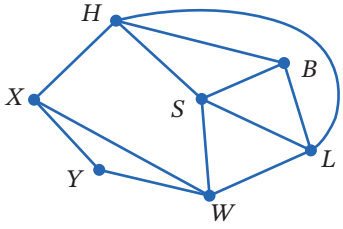
b



c



d

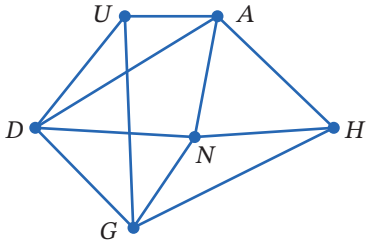


20 The graph shows how 6 towns are connected by a road network.

12.12

Is it possible to start at one town and go along each road once and finish at the same town?

Give a reason for your answer.



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